

Electrical and Electronic Engineering

ENEL700

COMMUNICATION ENGINEERING

Laboratory Manual



Auckland University of Technology

2026

Produced by Jack Xuejun Li

Copyright © 2026 by Jack Xuejun Li All Rights Reserved.

Department of Electrical and Electronic Engineering

School of Engineering, Computer and Mathematical Science

Auckland University of Technology

July 2026

Table of Contents

LAB 1	Amplitude Modulation	5
1	Objective	5
2	Theory	5
3	Method	7
4	Questions	8
LAB 2	Frequency Modulation	9
1	Objective	9
2	Theory	9
3	Problem.....	10
4	Method	10
5	Questions	11
LAB 3	Binary Phase-shift Keying.....	14
1	Objective	14
2	Theory	14
3	Experiment.....	16
4	Measurements	18
5	Questions	18
LAB 4	Pulse Coded Modulation	20
1	Objective	20
2	Theory	20
3	PCM Encoder Module	21
4	Experiment.....	22
5	Questions	24
6	PCM Decoding	25
7	The TIMS PCM Decoder Module	26
8	Experiment.....	27
LAB 5	Amplitude Modulation using Simulink	30
1	Objective	30
2	Theory	30
3	Equipment.....	30
4	Method and Questions	30
5	Questions	37
LAB 6	Harmonic analysis.....	39
1	Objective	39
2	Theory	39
3	Method	40
4	Questions	40
Appendix A	Receiver and Transmitter Block Diagrams.....	41
1	Modulator	42
2	Demodulator.....	43
3	Oscillator	43
4	Filter	44
5	Small-signal amplifier.	44
Appendix B	Waveform Spectra	46
1	Introduction	46
2	Complex Repetitive Waves	47

Guidelines

Each laboratory report must be typed. Each laboratory is related to material in the descriptor. Laboratory reports will be marked out of 20 with an average write-up receiving 12 marks. A penalty of 2 mark per day late will be imposed. Laboratory reports will not be accepted for marking if it is one week overdue. A well designed experiment can teach you a considerable amount about a topic.

The contents of the report must include the following components:

1. **Cover** – A standard cover is to be attached to the front of the report. This cover is available on Canvas. Please modify the cover, as per the instructions on the cover, and prepare one for each report. Save and use when required.
2. **Objective** – You may copy from the lab book.
3. **Theory** – it is worth your while expanding the theory section. You can use this as a set of quality notes on the topic of the experiment.
4. **Method** – If there is a change to the experimental method, you will have to document the method used. Otherwise, the comment “As per Laboratory book’ is all that is required.
5. **Results** – A complete documentation of the results that includes graphs, diagrams and supporting equations / mathematics. The graphs must be drawn on correctly scaled graph paper. If photographs are included, they must be in focus and easily read.
6. **Diagrams** must be neat and well labelled.
7. **Conclusion** - This section provides a place where the data may be fully discussed and interpreted. Here one may address questions like "why did something unexpected happen?", “why does that waveform look like that. It is not sufficient to comment that the experiment matched the aim.
8. **References** – You must make use of reference material – books, magazines, datasheets, internet etc. Get into the habit of not believing everything that you read, but improved your confidence by finding other references. List (in an approved fashion) all reference material.

Symbols

The graphical symbols for diagrams are specified in an *International Electrotechnical Commission (IEC)* publication and are recognised by *Standards NZ*. The library has a copy of the publication. IEC publication 617-10 *Graphical symbols for diagrams – part 10 – Telecommunication – Transmission*. 389.6 INT

I recommend that you have a look at these publications (there are a number of them) and familiarise yourself with the common symbols. Most textbooks do not conform to a standard.

Many larger companies (especially those from USA) tend to have their own symbols, which can cause confusion.

A block diagram is language independent and rapidly conveys the principles of a circuit. The diagram is the basis for many telecommunication topics.

LAB 1 AMPLITUDE MODULATION

1 Objective

To examine the characteristics of a monolithic balanced modulator.

2 Theory

The LM / MC1496 integrated circuit has become an industry standard and while old is still in production. It has been chosen for this experiment because it is versatile leading to a number of modulator configurations with minimal circuit changes.

The monolithic modulator functions as a broadband, double balanced, suppressed carrier balanced modulator without a requirement for transformers or tuned circuits. It is based on the Gilbert cell that is now universally used in modulator / mixer ICs.

In addition to its basic application as a balanced modulator (or demodulator) it can be used as an SSB product detector, AM modulator, AM demodulator, mixer, frequency doubler, phase detector. In short, it is a versatile integrated circuit.

Figure 1 shows a schematic diagram of the MC1496. For purposes of the analysis, the following conventional assumptions have been made for simplification:

1. Devices of similar geometry within a monolithic chip are assumed identical and matched where necessary, and
2. transistor base currents are ignored with respect to the magnitude of collector currents, therefore, collector and emitter currents are assumed equal.

Referring to Figure 1, the LM / MC1496 consists of a differential amplifier Q5–Q6 driving a dual differential amplifier composed of transistors Q1, Q2, Q3 and Q4.

Transistors Q7 and Q8 and associated bias circuitry form constant current sources for the lower differential amplifier Q5–Q6.

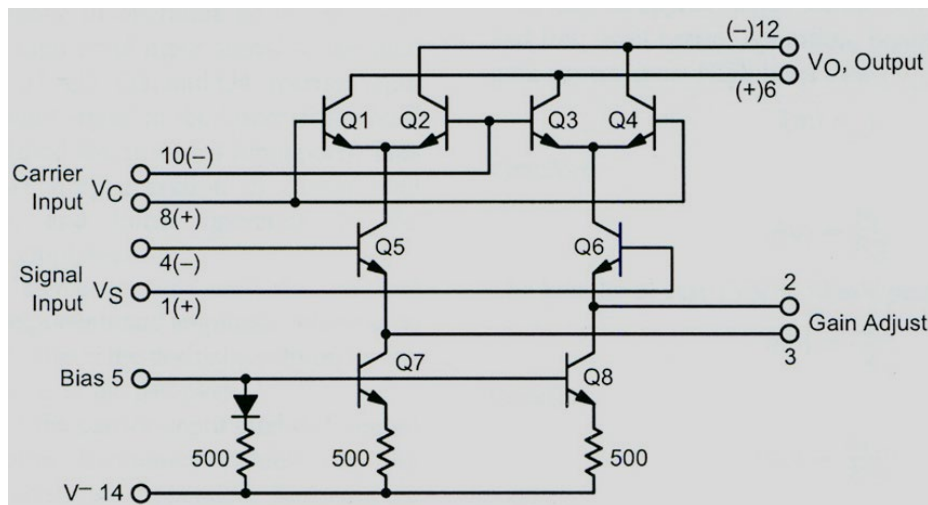


Figure 1: Schematic Diagram of LM/MC1496

The analysis of operation of the LM / MC1496 is based on the ability of the device to deliver an output which is proportional to the product of the input voltages V_C

and V_S . This holds true when the magnitudes of V_C and V_S are maintained within the limits of linear operation of the three differential amplifiers in the device.

Expressed mathematically, the output voltage (actually output current, which is converted to an output voltage by an external load resistance), V_O is given by –

$$V_O = KV_C V_S \quad (1)$$

where the constant K may be adjusted by choice of external components.

2.1 Device Operation

The most common mode of operation of the MC1496 consists of applying a high level input signal to the dual differential amplifiers, Q1, Q2, Q3, and Q4, (carrier input port) and a low level input signal to the lower differential amplifier, Q5 and Q6, (modulating signal input port).

This results in saturated switching operation of carrier dual differential amplifiers, and linear operation of the modulating differential amplifier.

The resulting output signal contains only the sum and difference frequency components and amplitude information of the modulating signal. This is the desired condition for the majority of the applications of the MC1496.

Saturated operation of the carrier–input dual differential amplifiers also generates harmonics (which may be predicted by Fourier analysis). Reducing the carrier input amplitude to its linear range greatly reduces these harmonics in the output signal.

However, it has the disadvantages of reducing device gain and causing the output signal to contain carrier signal amplitude variations.

The carrier input differential amplifiers have no emitter degeneration. Therefore, the carrier input levels for linear and saturated operation are readily calculated.

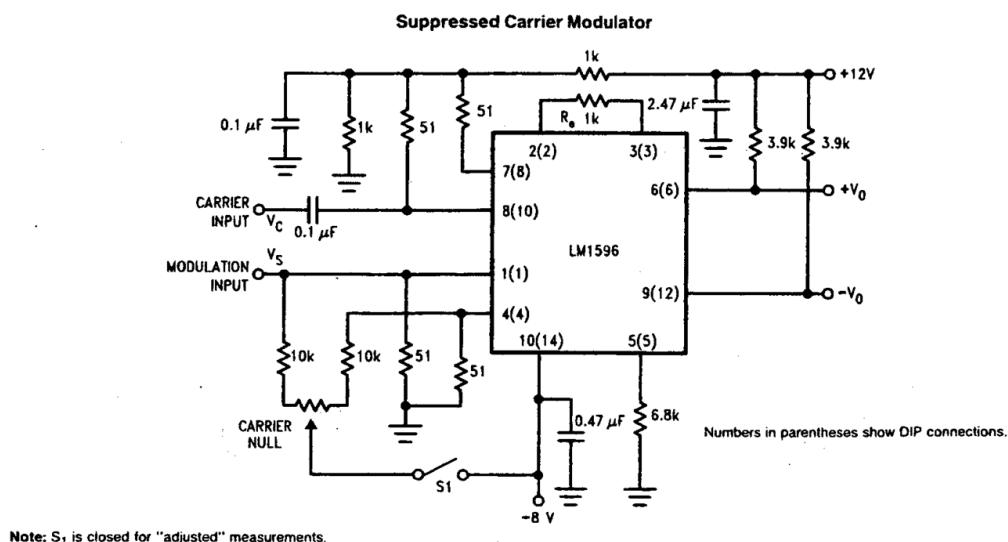


Figure 2: Suppressed Carrier Modulator

The crossover point is in the vicinity of 15–20 mV rms, with linear operation below this level and saturated operation above it. In this experiment, it is being operated in the saturated mode.

The modulating–signal differential amplifier has its emitters brought out to pins 2 and 3. This permits the designer to select his own value of emitter degeneration resistance and thereby tailor the linear dynamic range of the modulating–signal input port to a particular requirement. The resistor also determines device gain.

The balanced modulator circuit will function as an amplitude modulator if the carrier null circuit is unbalanced to insert the desired amount of carrier into the output signal. The circuit shown in Figure 2 allows modulation at any percentage from 0% to 100%.

3 Method

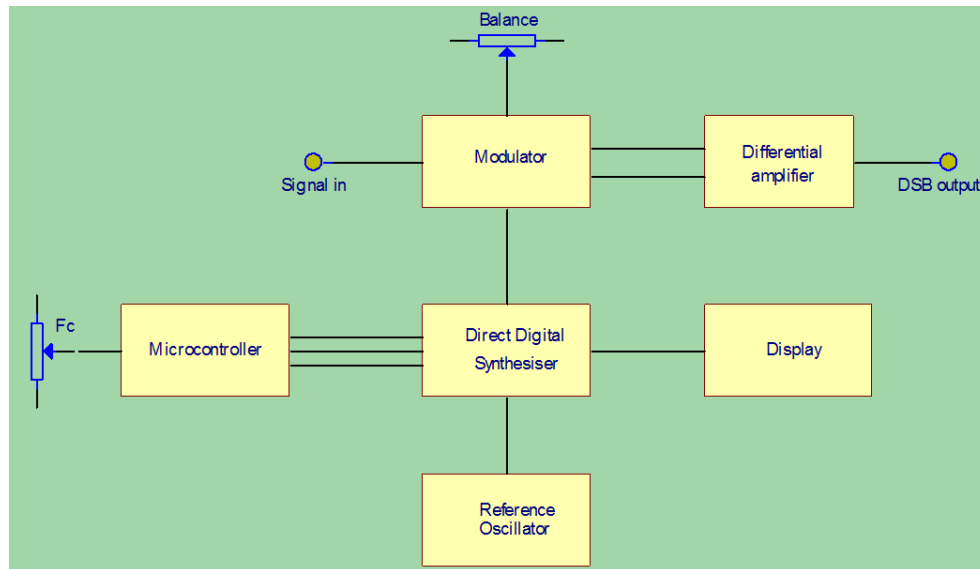


Figure 3: DSB generation circuit using monolithic modulator

- (a). Set up the circuit as shown in Figure 3. The power supply voltage is not to exceed ± 12 volts - polarity as indicated.
An internal carrier generator provides a signal around 455 kHz to the modulator. The frequency of this oscillator can be adjusted by approximately 20 kHz.
An external audio generator – Thurlby Tandar – is set to give a 5kHz sinusoidal signal. Set the level to about 100 mV as shown on the generator's meter. Connect the main output to the modulators signal in. This is the modulating signal. If the signal in is too large, then the DSB output will contain more unwanted signals.
- (b). The HP 8591E Spectrum Analyser is to be used to view the output in the frequency domain. Set the frequency to 450 kHz and the span to 50 kHz.
An Oscilloscope can be used in the time domain. Set the time-base to about 50 $\mu\text{s}/\text{div}$. Triggering will be difficult since the DSB out signal is a switched waveform of changing amplitude.
A Tee connector can be used to connect the two instruments together to the DSB output of the modulator.

3.1 BALANCED MODULATOR:

- (c). The circuit is to be balanced to suppress the carrier at the output. This requires the circuit to be set up as described above. The level of the audio signal may have to be adjusted to get a low level of distortion.
- (d). With the modulating signal (signal in) disconnected, adjust the balance potentiometer to reduce the amplitude of the carrier as seen on the spectrum analyser connected to the output of the modulator.

- (e). Reconnect the modulating signal. Observe and record the output waveform displayed on the Oscilloscope and the Analyser.
- (f). Using the Analyser, measure the amplitude of all output products for the modulator. (look for f_m , and multiples of f_c , and $f_c + f_m$, $f_c - f_m$) and plot on graph paper as a frequency domain diagram.

3.2 AMPLITUDE MODULATOR

- (g). Use the circuit configuration and levels as in the previous section.
- (h). Set the balance control fully clockwise to unbalance the modulator.
- (i). Record the output waveform as displayed on the Oscilloscope and the Analyser.
- (j). Measure the amplitude and the frequency of all the output products of the modulator and plot on graph paper as a frequency domain diagram.
- (k). Vary the level of the modulating signal and observe the change in the depth of modulation.

3.3 SSB GENERATION

- (l). Use the spectrum analyser to sweep and store the response of a 455 kHz ceramic band-pass filter. A centre frequency of 455 kHz and a span of 50 kHz should be sufficient.
- (m). To the output of the modulator, connect the BPF. Organise the display of the spectrum analyser to show the stored filter sweep with the output from the modulator superimposed on top. Adjust the carrier frequency, in turn, to the low side of the filter response and then the high side of the filter response. Observe and record the frequency-domain and the time-domain signals at the output of the filter.

4 Questions

- Question 1. How effective is this modulator as an AM and as a DSBSC modulator?
- Question 2. Define the term modulation index
- Question 3. Calculate the modulation index of the AM waveform that you have recorded in part B.
- Question 4. What are the advantages and the disadvantages of the monolithic modulator in comparison with the diode modulator?
- Question 5. What is the effect, in both the time domain and frequency domain, of varying the frequency of the modulating signal?
- Question 6. What is the effect, in both the time domain and frequency domain, of varying the frequency of the carrier?
- Question 7. Assume that the response of the SSB band-pass filter extends from 128 KHz to 132 kHz and is of a quality in that the slope of the sides of the filter response is very steep (close to vertical). If the modulating signal is a 1 kHz sinusoidal signal and the carrier is 132 kHz, what signal (in the time-domain) will be present at the output of the filter?

LAB 2 FREQUENCY MODULATION

1 Objective

To investigate the spectral content of a FM waveform.

2 Theory

The frequency modulated waveform is a *complex* waveform. An complex waveform consists of more than one sinusoidal waveform whose frequency and amplitude determine the shape of the complex waveform.

When a sinusoidal carrier wave of frequency f_c is frequency modulated by a sinusoidal signal of frequency f_m , the modulated wave may contain components at a number of different frequencies. These frequencies are the carrier frequency and a number of side-frequencies positioned either side of the carrier frequency. The side-frequencies are spaced apart at frequency intervals equal to the modulating frequency. The first order side-frequencies are $f_c \pm f_m$, the second order side-frequencies are $f_c \pm 2f_m$, the third order side-frequencies are $f_c \pm 3f_m$ and so on.

The amplitude of the various components, including the carrier, depends upon the modulation index β or the deviation ratio D . The diagram below, is a graph of Bessel's functions that give the amplitude of the various frequencies.

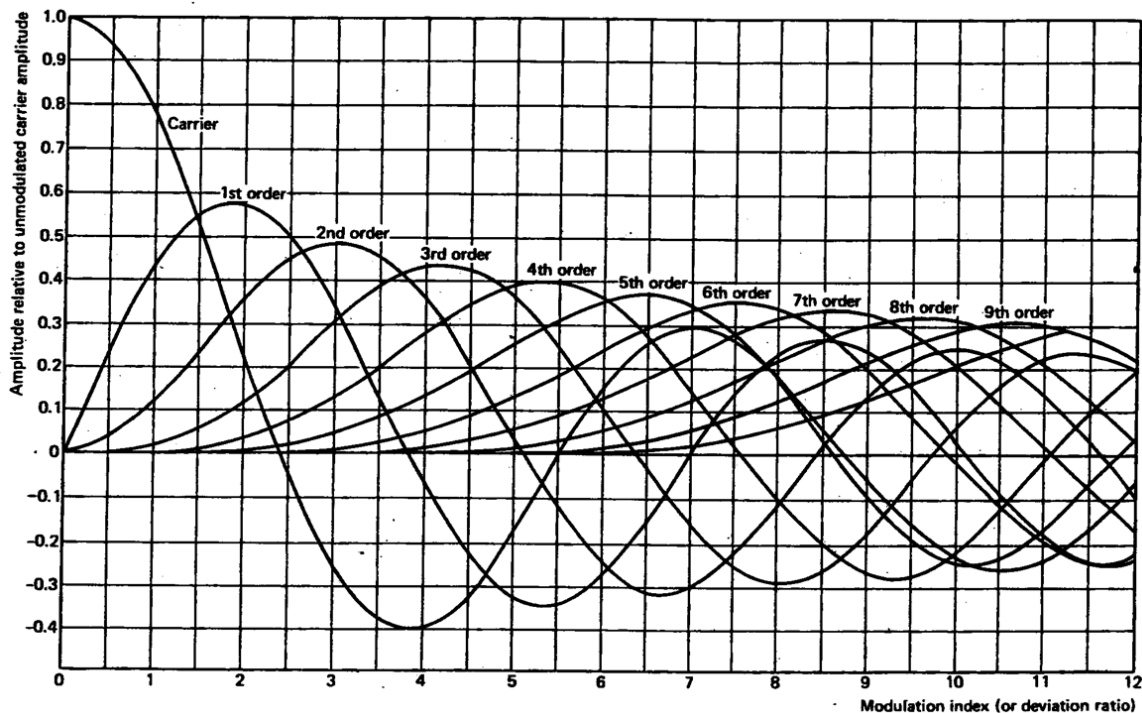


Figure 4: Relative amplitudes of different frequencies

Only the first nine orders of side-frequencies have been shown in order to clarify the drawing but many more are equally possible. Projecting from the modulation index value onto the appropriate curve, and then onto the vertical axis obtains the amplitudes of each component present in the wave. Negative signs are omitted since only the magnitude of each component is wanted.

For example, a modulation index of 5 produces the following amplitudes.

Carrier	- 0.18	1 st order	- 0.33
2 nd order	- 0.05	3 rd order	- 0.36
4 th order	- 0.39	5 th order	- 0.26

and so on. These values are relative to an unmodulated carrier amplitude of 1.

A spectrum diagram can then be drawn.

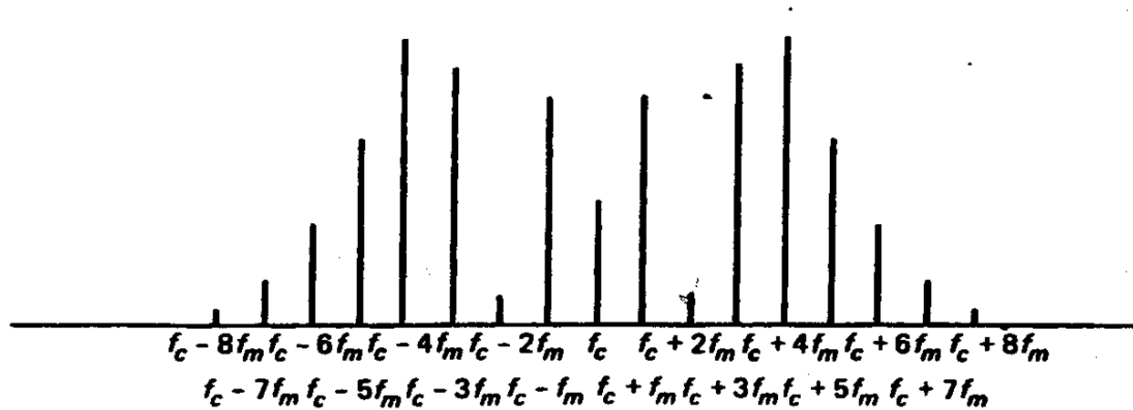


Figure 5: Amplitude spectrum of FM

The modulation index is defined as the ratio of the frequency deviation of the carrier to the modulating signal frequency.

$$\beta = f_d / f_m \quad (2)$$

3 Problem

A 100 MHz carrier is modulated by a 15 kHz sine wave. If the modulation index is 8, plot the spectrum diagram. Ensure that the values of the amplitude and the frequencies are shown on the diagram.

A spectrum analyser is an instrument that can be used to observe the spectrum of the frequency modulated waveform. Most high quality RF signal generators will produce a frequency modulated waveform. Note that

1. The signal generators in WS414 have the ability to produce a modulated waveform. The accuracy of the waveform depends on the quality of the oscillator (and therefore the cost).
2. Both the Agilent 33522A Arbitrary waveform generator and the Rohde & Schwarz SMC100A RF signal generator produce excellent results though they are both a little hard to use.

EQUIPMENT: Agilent 33522A arbitrary waveform generator, HP ESA-1500A spectrum analyser, connecting cable.

4 Method

- (a). Instrument set-up –
Signal generator – Pressing the appropriate key will bring up soft-menus which can have several levels. Values can be entered using the keypad (best) or the rotary knob.

Connect the coaxial cable to output 2 (green) and then to the input of the spectrum analyser. Output 2 is used because it is less obstructive than output 1. Note that it has a 50Ω output impedance which matches the input impedance of the spectrum analyser.

Waveforms – sine, frequency 10 MHz, amplitude -20dBm, offset and phase zero.

Modulate – modulate on, type FM, source internal, frequency deviation 5 kHz, shape sine, FM frequency 5 kHz.

Channel setup 2 – output load 50Ω, output on.

The generator should now be producing a modulated waveform with its parameters displayed on screen. Press the modulate button as this is the parameters that you will change.

The frequency deviation can be changed when required.

Spectrum analyser – Again, pressing the appropriate buttons will activate a set of on screen soft-menus. Values can be changed using the keypad and the rotary knob.

Frequency - set the centre frequency to 10 MHz.

Span - set the span to 50 kHz. Since there are 10 major horizontal divisions, each division represents 5 kHz.

Amplitude-The vertical scale has 10 major divisions. Relative measurements will be made, relative to the full scale which can represent 100% or a value of 1.

Press the Amplitude button and set the Scale type to LIN. Use the rotary knob to adjust the sensitivity so that the amplitude of the unmodulated carrier (modulate off) just touches the top line of the vertical scale. The vertical scale can now be read relative to 1 (or 100%) at the top of the scale. A signal that has an amplitude of 5 divisions up from the bottom will have an amplitude of 0.5 (or 50%).

- (b). Turn the modulation on (modulate on), and for the following values of frequency deviation record the frequencies and amplitudes of the carrier and the side-frequencies.

- (i) Deviation = 5 kHz,
- (ii) Deviation = 10 kHz,
- (iii) Deviation = 15 kHz.

- (c). Determine the modulation index for the following occurrences –

- (i) the first carrier null (zero amplitude),
- (ii) the 1st order side-frequency null,
- (iii) the 2nd order side-frequency null.

5 Questions

Question 1. Calculate the modulation index for the three cases in Method 2.

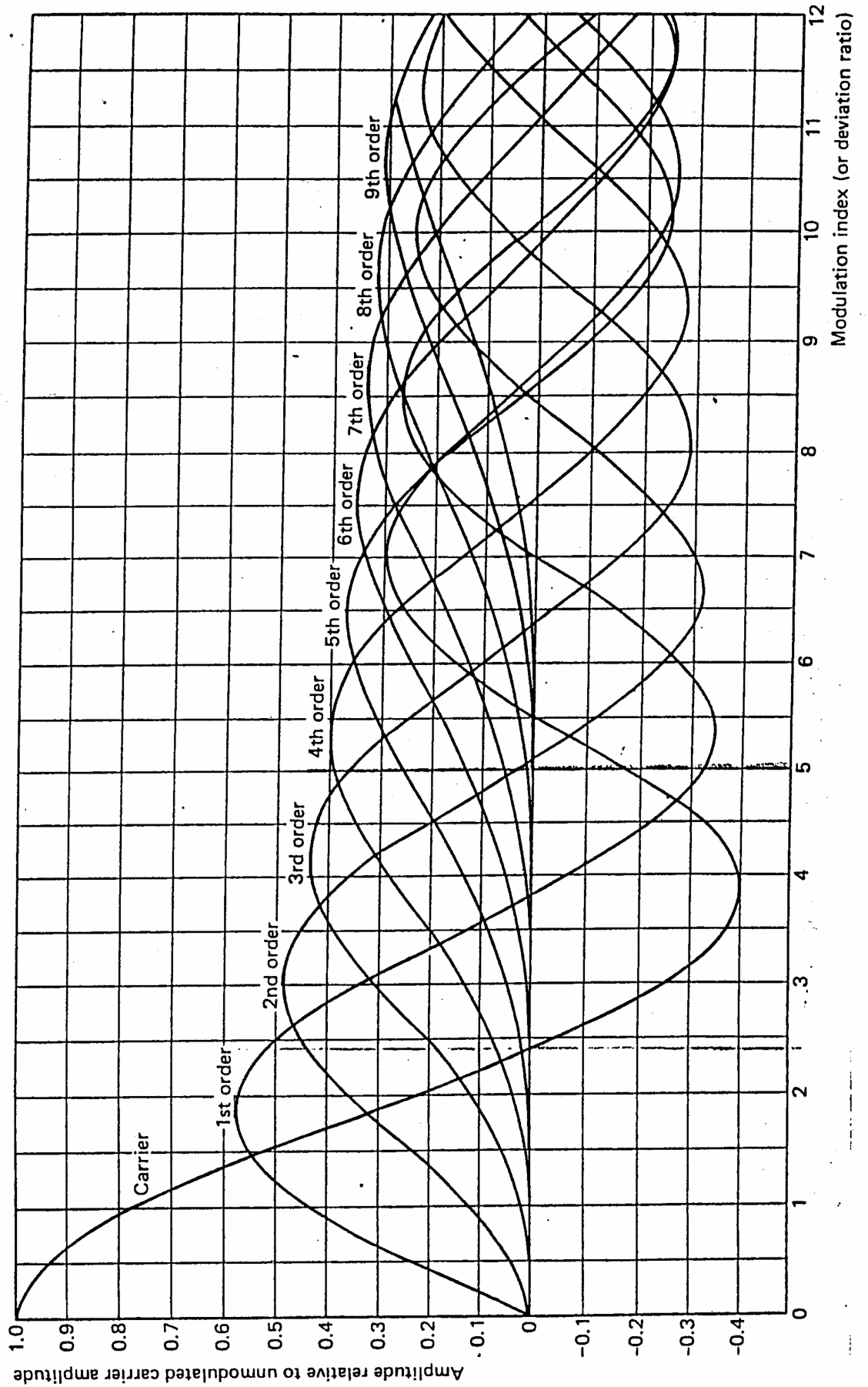
Question 2. What would cause a discrepancy between the measured amplitudes and the amplitude of the carrier and side-frequency in the Bessel's function chart.

Question 3. For the three cases in Method 3, determine the normalised power of the frequency modulated wave.

Question 4. The formula is $J_0^2(\beta) + 2 \sum_{n=1}^{\infty} J_n^2(\beta) = 1$. A spreadsheet may be the easiest way to calculate these values.

Question 5. Explain any variations between the expected value and measured value of the normalised power.

FREQUENCY MODULATION 23



The relative amplitudes of the side-frequencies (Bessel functions) rounded to 2 decimal places. Values less than 0.01 (1%) are entered as zero.

Table I Bessel Function Table

Modulation index (β)													
	0	1	2	3	4	5	6	7	8	9	10	11	12
J_0	1.0	0.77	0.22	0.26	0.40	0.18	0.15	0.30	0.17	0.09	0.25	0.17	0.05
J_1		0.44	0.58	0.34	0.07	0.33	0.28	0.00	0.23	0.25	0.04	0.18	0.22
J_2		0.11	0.35	0.49	0.36	0.05	0.24	0.30	0.11	0.14	0.25	0.14	0.08
J_3		0.02	0.13	0.31	0.43	0.36	0.11	0.17	0.29	0.18	0.06	0.23	0.20
J_4			0.03	0.13	0.28	0.39	0.36	0.16	0.11	0.27	0.22	0.02	0.18
J_5			0.01	0.04	0.13	0.26	0.36	0.35	0.19	0.06	0.23	0.24	0.07
J_6				0.01	0.05	0.13	0.25	0.34	0.34	0.20	0.01	0.20	0.24
J_7					0.02	0.05	0.13	0.23	0.32	0.33	0.22	0.02	0.17
J_8						0.02	0.06	0.13	0.22	0.31	0.32	0.22	0.05
J_9						0.01	0.02	0.06	0.13	0.21	0.29	0.31	0.23
J_{10}							0.01	0.02	0.06	0.12	0.21	0.28	0.30
J_{11}								0.01	0.03	0.06	0.12	0.20	0.27
J_{12}									0.01	0.03	0.06	0.12	0.19
J_{13}										0.01	0.03	0.06	0.12
J_{14}											0.01	0.03	0.07
J_{15}												0.01	0.03
J_{16}													0.01

Side-frequency pairs

Note: Blank cell have a value of zero.
 J_0 is the carrier.

For a better table, check URL:

<https://www.yumpu.com/en/document/read/39029881/bessel-function-table> .

LAB 3 BINARY PHASE-SHIFT KEYING

1 Objective

To use the Telecommunication Instructional Modelling System (TIMS) to investigate a digitally modulated signal that uses Binary Phase-shift keying.

2 Theory

2.1 Generation of BPSK

Consider a sinusoidal carrier. If it is modulated by a bi-polar bit stream according to the scheme illustrated in Figure 6 below, its polarity will be reversed every time the bit stream changes polarity. This, for a sine-wave, is equivalent to a phase reversal (shift). The multiplier output is a BPSK signal.

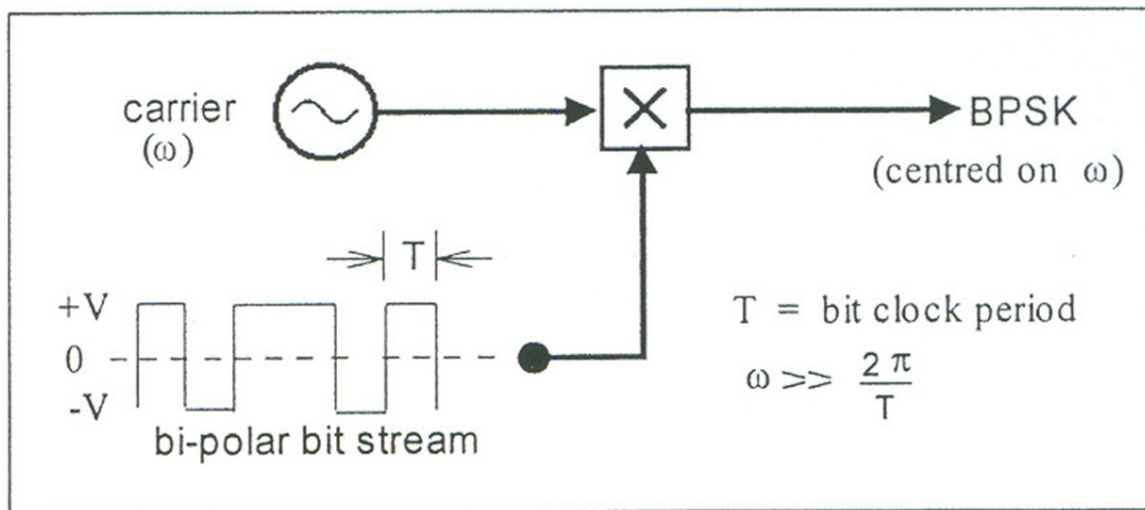


Figure 6: Generation of BPSK

The information about the bit stream is contained in the changes of phase of the transmitted signal.

A synchronous demodulator would be sensitive to these phase reversals.

The appearance of a BPSK signal in the time domain is shown in Figure 7 (lower trace). The upper trace is the binary message sequence.

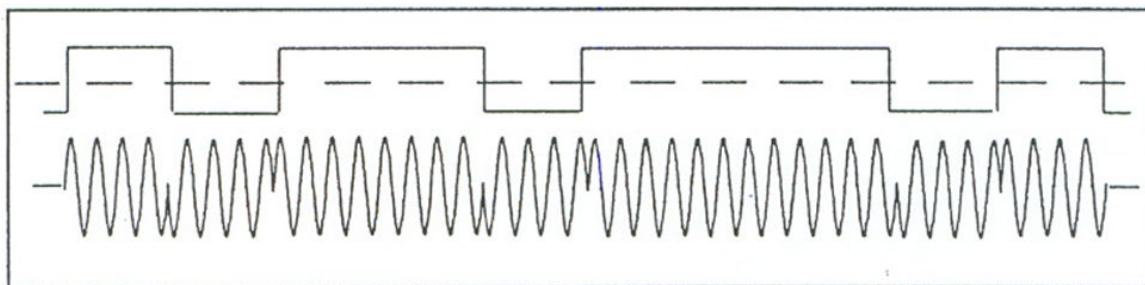


Figure 7: A BPSK signal in the time domain.

There is something special about the waveform of Figure 7. The wave shape is 'symmetrical' at each phase transition. This is because the bit rate is a sub-multiple of the carrier frequency $\omega/2\pi$. In addition, the message transitions have been timed to occur at a zero-crossing of the carrier.

Whilst this is referred to as 'special', it is not uncommon in practice. It offers the advantage of simplifying the bit clock recovery from a received signal. Once the carrier has been acquired then the bit clock can be derived by division.

But what does it do to the bandwidth? See Question 4.

2.2 Band-limiting

The basic BPSK generated by the simplified arrangement illustrated in Figure 6 will have a bandwidth in excess of that considered acceptable for efficient communications.

If you can calculate the spectrum of the binary sequence then you know the bandwidth of the BPSK itself. The BPSK signal is a *linearly modulated* DSB, and so it has a bandwidth twice that of the base-band data signal from which it is derived (assuming $\omega > 2B$).

In practice there would need to be some form of bandwidth control.

Band-limiting can be performed either at base-band or at carrier frequency. It will be performed at base-band in this experiment.

2.3 BPSK demodulation

Demodulation of a BPSK signal can be considered a two-stage process.

1. Translation back to base-band, with recovery of the band-limited message waveform

Regeneration from the band limited waveform back to the binary message bit stream.

Translation back to base-band requires a local, synchronized carrier.

2.3.1 Stage 1

Translation back to base-band is achieved with a synchronous demodulator, as shown in Figure 8 below.

This requires a local synchronous carrier. In this experiment a stolen carrier will be used.

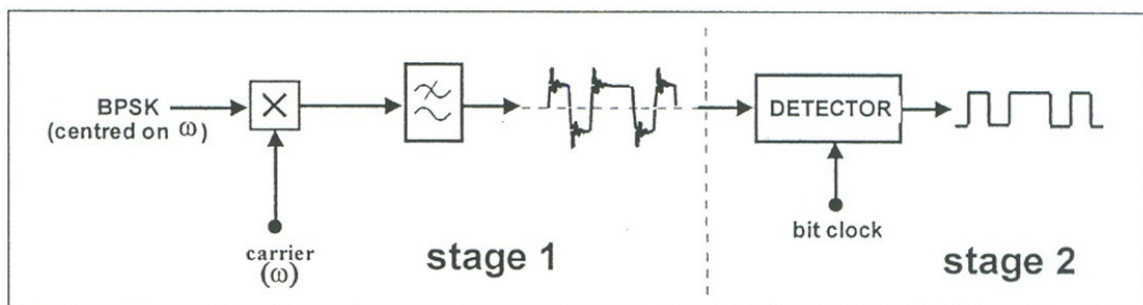


Figure 8: Synchronous demodulation of BPSK

2.3.2 Stage 2

The translation process does not reproduce the original binary sequence, but a band-limited version of it.

The original binary sequence can be regenerated with a detector. This requires information regarding the bit clock rate. If the bit rate is a sub-multiple of the carrier frequency then bit clock regeneration is simplified.

In TIMS the DECISION MAKER module can be used for the regenerator, and in this experiment the bit clock *will* be a sub-multiple of the carrier.

2.3.3 Phase Ambiguity

You will see in the experiment that the sign of the phase of the demodulator carrier is important.

Phase ambiguity is a problem in the demodulation of a BPSK signal.

There are techniques available to overcome this. One such sends a training sequence, of known format, to enable the receiver to select the desired phase, following which the training sequence is replaced by the normal data (until synchronism is lost!).

An alternative technique is to use differential encoding. This will be demonstrated in this experiment by selecting a different code from the LINE-CODE ENCODER.

3 Experiment

3.1 The BPSK generator

The BPSK generator of Figure 6 is shown in expanded form in Figure 9, and modelled in Figure 5

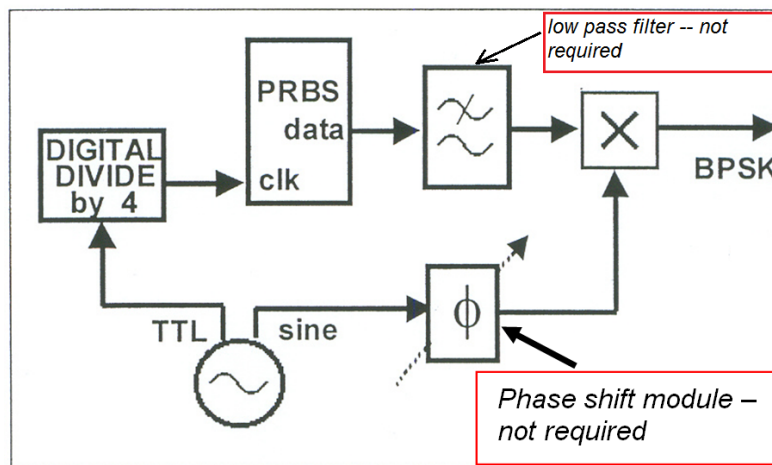


Figure 9: Block diagram of BPSK generator to be modelled

Note that the carrier will be four times the bit clock rate.

The low-pass filter is included as a band limiter if required. Alternatively a band pass filter could have been inserted at the output of the generator. Being a linear system, the effect would be the same.

The AUDIO OSCILLATOR supplies a TTL signal for the bit clock digital DIVIDE-BY-FOUR sub-system in the LINE-CODE ENCODER, and a sinusoidal signal for the carrier.

The digital DIVIDE-BY-FOUR sub-system within the LINE-CODE ENCODER is used for deriving the bit clock as a sub-multiple of the BPSK carrier. Because the

DECISION MAKER, used in the receiver, needs to operate in the range about 2 to 4 kHz, the BPSK carrier will be in the range about 8 to 16Hz.

The NRZ-L code is selected from LINE-CODE ENCODER.

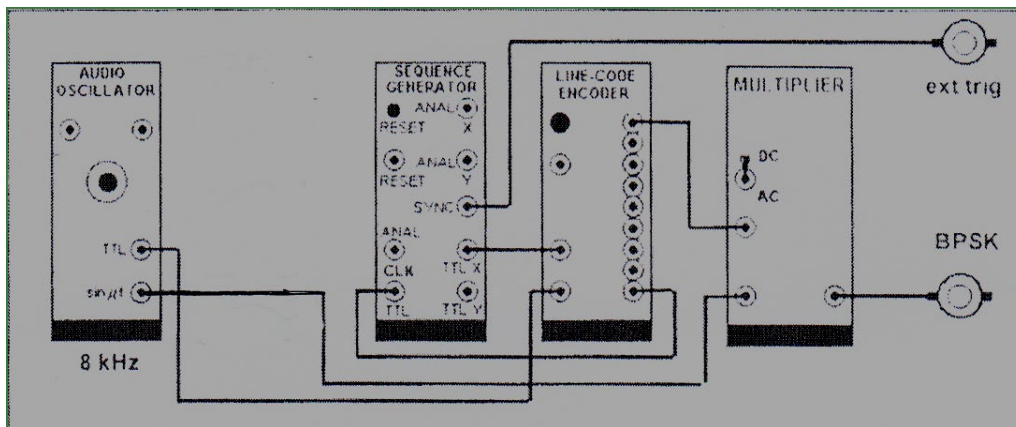


Figure 10: Model of the BPSK generator

Viewing of the phase reversals of the carrier is simplified because the carrier and binary clock frequencies are harmonically related.

- (d). Patch up the modulator of Figure 10. Acquaint yourself with a BPSK signal. Examine the transitions as the phase between bit clock and carrier is altered. Vary the bandwidth of the PRBS with the TUNEABLE LPF. Notice the envelope.

3.2 BPSK demodulator

Figure 3 shows a synchronous demodulator for a BPSK signal in block diagram form. This has been modelled in Figure 11 below. In the first part of the experiment the carrier and bit clocks will be stolen.

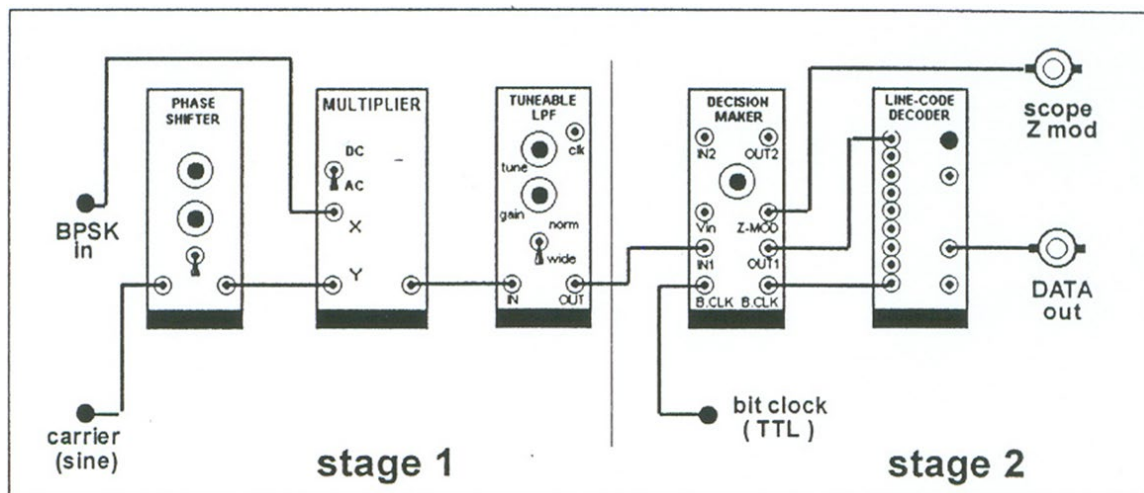


Figure 11: The BPSK demodulator

The phase of the carrier is adjustable with the PHASE SHIFTER for maximum output from the low-pass filter. Phase reversals of 180° can be introduced with the front panel toggle switch.

Select the NRZ-L input to the LINE-CODE DECODER. The LINE-CODE ENCODER and LINE-CODE DECODER modules are not essential in terms of the coding they introduce (since a bi-polar sequence is already available from the

SEQUENCE GENERATOR) but they are useful in that they contain the DIVIDE-BY-FOUR sub-systems, which are used to derive the sub-multiple bit clock.

The LPF following the demodulator multiplier is there to remove the components at double the carrier frequency. Its bandwidth can be set to about 12 kHz, although, for maximum signal-to-noise ratio (if measuring bit error rates, for example), something lower would probably be preferred.

4 Measurements

The BPSK will have been band-limited by the low-pass filter in the transmitter, and so the received waveform is no longer rectangular in shape. But you can observe that the demodulator filter output is related to the transmitted sequence (the NRZ-L code introduces only a level shift and amplitude scale).

The DECISION MAKER will regenerate the original TTL sequence waveform.

Notice the effect upon the recovered sequence when the carrier phase is reversed at the demodulator.

The following Tasks are a reminder of what you might investigate.

- (e). Patch up the demodulator of Figure 11. The received signal will have come from the transmitter of Figure 10. Observe the output from the TUNEABLE LPF, and confirm its appearance with respect to that transmitted. If the sequence is inverted then toggle the front panel 180° switch of the receiver PHASE CHANGER.
- (f). Set the on-board switch SW1 of the DECISION MAKER to accept NRZ-L coding. Use the gain control of the TUNEABLE LPF to set the input at about the TMS ANALOG REFERENCE LEVEL of ± 2 volt peak. Adjust the decision point. Check the output.
- (g). Observe the TTL output from the LINE-CODE DECODER. Confirm that the phase of the receiver carrier (for the NRZ-L line code) is still important.
- (h). Investigate a change of bandwidth of the transmitted signal. Notice that, as the bandwidth is changed, the amplitude of the demodulated sequence at the DECISION MAKER input will change. This you might expect, but, under certain conditions, it can increase as the bandwidth is decreased! How could this be? See Question 6.

5 Questions

- Question 1. Do you think BPSK is an analogue signal? Any comments?
- Question 2. In the model of Figure 10, is it necessary that the MULTIPLIER be switched to DC, as shown?
- Question 3. You observed the shape of the phase transitions as the PHASE SHIFTER of Figure 11 was changed. Would this influence the spectrum of the BPSK signal?
- Question 4. Does making the bit rate a sub-multiple of the carrier frequency have any influence on the spectrum of the BPSK signal?
- Question 5. What is the purpose of the low-pass filter in the BPSK demodulator model? What determines its bandwidth?
- Question 6. The amplitude of the signal at the DECISION MAKER input can decrease as the bandwidth of the transmitter is widened (or vice versa). At first glance this seems unusual? Explain.

Question 7. *The PHASE SHIFTER in the demodulator of Figure 11 was adjusted for maximum output. What phase was it optimizing, and what was the magnitude of this phase? Could you measure it?*

LAB 4 PULSE CODED MODULATION

1 Objective

To use the Telecommunication Instructional Modelling System (TIMS) to investigate pulse coded modulation.

2 Theory

This is an introductory experiment into pulse code modulation (PCM) and consists of two parts:

1. PCM encoder - This module generates a PCM output signal from an analogue input message and,
2. PCM decoder – This module takes the PCM signal and recreates the analogue signal.

2.1 PCM ENCODING

The input to the PCM ENCODER module is an analogue message. This must be constrained to a defined bandwidth and amplitude range.

The maximum allowable message bandwidth will depend upon the sampling rate to be used. The Nyquist criterion must be observed.

The amplitude range must be held within a 2.0 volts (peak) range.

This is in keeping with the input amplitude limits set for all analogue modules.

A step-by-step description of the operation of the module follows:

1. The module is driven by an external TTL clock.
2. The input analogue message is *sampled* periodically. The *sample rate* is determined by the external clock.
3. The sampling is a *sample-and-hold* operation. It is internal to the module, and cannot be viewed by the user. What is held is the *amplitude* of the analogue message *at the sampling instant*.
4. Each sample amplitude is compared with a finite set of amplitude levels. These are distributed (uniformly, for *linear* sampling) within the range ± 2.0 volts. These are the system *quantising* levels.
5. Each quantising level is assigned a *number*, starting from zero for the lowest (most negative) level, with the highest number being (L-1), where L is the available number of levels. For an 'n' bit system, the number of levels (L) will be 2^n .
6. Each sample is *assigned* a digital (binary) code word representing the number associated with the quantising level which is closest to the sample amplitude. The number of bits 'n' in the digital code word will depend upon the number of quantising levels. In fact, $n = \log_2 L$.
7. The code word is *assembled into a time frame* together with other bits as may be required (described below). In the TIMS PCM ENCODER (and many commercial systems) a single extra bit is added, in the least significant bit position. This is alternately a *one* or a *zero*. These bits are used by subsequent decoders for frame synchronisation.

8. The *frames* are transmitted serially. They are transmitted at the same rate as the samples are taken (but see I Question 3). The serial bit stream appears at the output of the module.
9. Also available from the module is a synchronising signal FS ('frame synch'). This signals the *end* of each data frame.

3 PCM Encoder Module

3.1 Front Panel Features

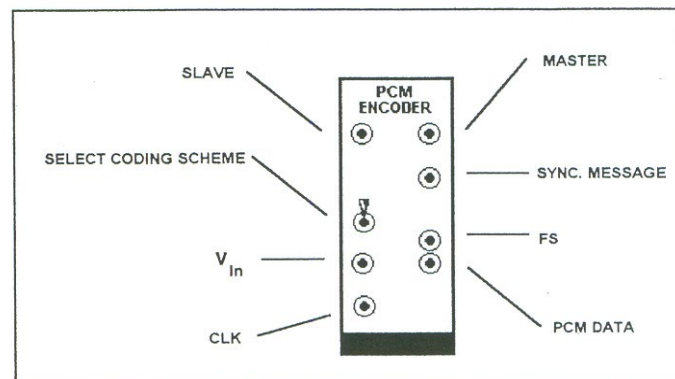


Figure 12: front panel layout of the PCM ENCODER

The front panel layout of the module is shown in Figure 12. Technical details are described in the *TIMS Advanced Modules User Manual*.

The input / output connections, and the three-position toggle switch use are:

1. **SELECT CODING SCHEME:** a three-position toggle switch which selects the 4-bit or 7-bit encoding scheme of the analogue samples.
2. **FS:** frame synchronisation, a signal which indicates the end of each data frame.
3. **V_{in}:** the analogue signal to be encoded.
4. **PCM DATA:** the output data stream, the examination of which forms the major part of this experiment.
5. **CLK:** this is a TIL (red) input, and serves as the MASTER CLOCK for the module. For this experiment you will use the 8.333 kHz TIL signal from the MASTER SIGNALS module.

3.2 TIMS PCM time frame

Each binary word is located in a *time frame*. The time frame contains eight *slots* of equal length, and is eight clock periods long. The slots, from first to last, are numbered 7 through 0. These slots contain the bits of a binary word. The least significant bit (LSB) is contained in slot 0.

The LSB consists of alternating *ones* and *zeroes*. These are placed ('embedded') in the frame by the encoder itself, and cannot be modified by the user. They are used by subsequent decoders to determine the location of each frame in the data stream, and its length.

The remaining seven slots are available for the bits of the binary code word. Thus the system is capable of a resolution of seven-bits maximum. This resolution, for purposes of experiment, can be reduced to four bits (by front panel switch). The 4-bit mode uses only five of the available eight slots - one for the embedded frame

synchronisation bits, and the remaining four for the binary code word (in slots 4, 3, 2, and 1).

3.3 Pre-calculations

You will be using an 8.333 kHz master clock. Answer the following question now, *before* commencing the experiment.

Question 1. From your knowledge of the 'PCM ENCODER module, obtained during preparation for the experiment, calculate the sampling rate of the analogue input signal. Show that it is the same for both the 4-bit and the 7-bit coding schemes. What can you say about the bandwidth of an input analogue signal to be encoded?

4 Experiment

The only module required for this experiment is a TIMS PCM ENCODER. Connecting-up the module.

To determine some of the properties of the analogue-to-digital conversion process it is best to start with a DC message. This ensures completely stable oscilloscope displays, and enables easy identification of the quantising levels.

Selecting the 4-bit encoding scheme, reduces the number of levels (2^4) to be examined.

- (a). Insert the module into the TIMS frame. Switch the front panel toggle switch to 4-BIT LINEAR (i.e., no companding).
- (b). Patch the 8.333 kHz TTL SAMPLE CLOCK from the MASTER SIGNALS module to the CLK input of the PCM ENCODER module.
- (c). Connect the V_{in} input socket to ground of the variable DC module.
- (d). Connect the frame synchronisation signal FS to the oscilloscope external synch. input.
- (e). On CH 1 display the frame synchronisation signal FS. Adjust the sweep speed to show three frame markers. These mark the end of each frame.
- (f). On CH 2 display the CLK signal.
- (g). Record the number of clock periods per frame.

Currently the analogue input signal is zero volts (V_{in} is grounded). Before checking with the oscilloscope, consider what the PCM output signal might look like. Make a sketch of this signal, fully annotated. Then:

- (h). On CH 2 display the PCM DATA from the PCM DATA output socket.

There *is* an analogue *input* signal to the encoder. It is of zero volts. This will have been coded into a 4-bit binary *output* number, which will appear in *each* frame. It depends on the coding scheme used and need not be '0000'. The *same* number appears in *each* frame because the analogue input is *constant*.

Your display should be similar to that of Figure 13 below, except that this shows five frames (too many frames on the oscilloscope display makes bit identification more difficult).

Knowing:

- 1. the number of slots per frame is 8

2. the location of the least significant bit is coincident with the end of the frame
3. the binary word length is four bits
4. the first three slots are 'empty' (in fact filled with zeros, but these remain unchanged under all conditions of the 4-bit coding scheme)

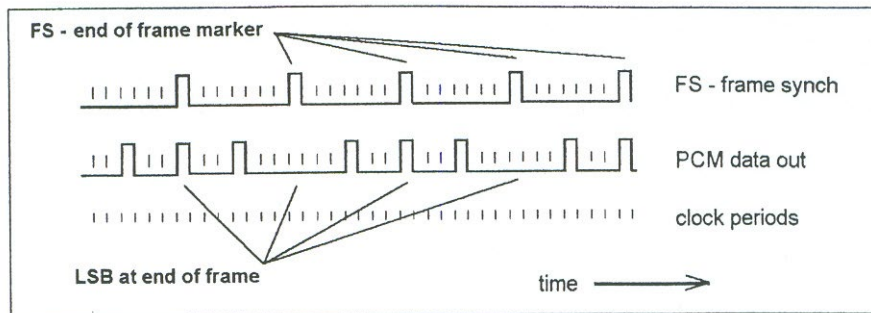


Figure 13: 5 frames of 4-bit PCM output for zero amplitude input

then:

- (i). Identify the binary word in slots 4, 3, 2, and 1.

4.1 Quantising levels for 4-bit linear encoding

You will now proceed to determine the quantising / encoding scheme for the 4-bit linear case.

- (j). Remove the ground connection, and connect the output of the VARIABLE DC module to V_{in} . Sweep the DC voltage slowly backwards and forwards over its complete range, and note how the data pattern changes in discrete jumps.
- (k). Use a DC voltmeter to monitor the DC amplitude at V_{in} . Adjust V_{in} to its maximum negative value. Record the DC voltage and the pattern of the 4-bit binary number.
- (l). Slowly increase the amplitude of the DC input signal until there is a sudden change to the PCM output signal format. Record the format of the new digital word, and the input amplitude at which the change occurred.
- (m). Continue this process over the full range of the DC supply.
- (n). Draw a graph showing the quantising levels, in volts, (on the horizontal) and their decimal equivalents of the associated binary numbers.

4.2 4-bit data format

From measurements made so far you should be able to answer the questions:

- Question 2.**
- (a) *What is the sampling rate?*
 - (b) *What is the frame width?*
 - (c) *What is the width of a data bit?*
 - (d) *What is the width of a data word?*
 - (e) *How many quantising levels are there?*
 - (f) *Are the quantising levels uniformly (linearly) spaced?*

4.3 Companding

This module is to be used in conjunction with the PCM DECODER in a later experiment. As a pair they have a *companding* option. There is compression in the encoder, and expansion in the decoder. In the encoder this means the

quantising levels are closer together for small input amplitudes - that is, in effect, that the input amplitude peaks are compressed during encoding. At the decoder the 'reverse action' is introduced to restore an approximate linear input/output characteristic.

It can be shown that this sort of characteristic offers certain advantages, especially when the message has a high peak-to-average amplitude characteristic, as does speech, and where the signal-to-noise ratio is not high.

This improvement will not be checked in this experiment. But the existence of the non-linear quantisation in the encoder will be confirmed.

With the *PCM decoder*, it will be possible to check the input/output linearity of the modules as a compatible pair.

- (o). Change to 4-bit companding by use of the front panel toggle switch.
- (p). The TIMS A4 companding law has already been selected by on-board switches. Make the necessary measurements to determine the nature of the law.

4.4 Periodic messages

A periodic 'synchronised' message is available at the Sync message terminal. Since it is synchronised to the master clock, it provides a stable oscilloscope display.

- (q). Take a periodic message from the SYNC. MESSAGE socket. Adjust the oscilloscope to display the message. Record its frequency and shape. Check if these are compatible with the Nyquist criterion; adjust the amplitude if necessary with one of the BUFFER AMPLIFIERS.
- (r). Now look at the PCM DATA output. Synchronise the oscilloscope (as previously) to the frame (FS) signal. Display two or three frames on CH 1, and the PCM DATA output on CH 2.

You will see that the data signal reveals very little. It consists of many overlaid digital words, all different.

One would need more sophisticated equipment (a digital analyser, a storage oscilloscope, ability to capture a single frame, and so on) to deduce the coding and quantising scheme from such an input signal.

5 Questions

- Question 3. *Define what is meant by the data 'frame' in this experiment. Draw a diagram showing the composition of a frame for the 4-bit coding scheme.*
- Question 4. *It is possible to transmit each frame at a much slower rate than it was produced (and, of course, recover the- original message as well). Explain how this might be done. When might this be an advantage?*
- Question 5. *Explain why a DC message gives a stable oscilloscope display of the PCM DATA output. Why is the display 'unstable' when a sine wave (for example) is the message?*
- Question 6. *For the 4-bit encoder draw a diagram showing the amplitude quantisation levels and the corresponding binary numbers used to encode them. Describe how this information was obtained experimentally.*

6 PCM Decoding

6.1 Signal source

The signal to be decoded will be provided by you, using PCM ENCODER module as set up earlier in the experiment.

6.2 Clock synchronisation

A clock synchronisation signal will be stolen from the encoder.

6.3 Frame synchronisation

In the PCM DECODER module there is circuitry which automatically identifies location of each frame in the serial data stream. To do this it collects groups of eight data bits and looks for the repeating pattern of alternate ones and zeros placed there (embedded) by the PCM ENCODER in the LSB position.

It can be shown that such a pattern cannot occur elsewhere in the data stream provided that the original band-limited analogue signal is sampled at or below Nyquist rate.

When the embedded pattern is found an 'end-of-frame' synchronisation signal FS is generated, and made available at the front panel.

The search for the frame is continuously updated. Why?

6.4 Companding

You should prepare by reading Page 333 *Communication Engineering Principles* about the principles of *companding*. You will already be aware that the PCM ENCODER module can incorporate compression into its encoding scheme. The PCM DECODER module can introduce the complementary expansion.

6.5 PCM decoding

The PCM DECODER module is driven by an external clock. This clock signal is synchronised to that of the transmitter. For this experiment a 'stolen' clock will be used. The source of frame timing information has been discussed above.

Upon reception, the PCM DECODER:

1. Extracts a frame synchronisation signal FS from the data itself (from the embedded alternate ones and zeros in the LSB position), or uses an FS signal stolen from the transmitter (see above).
2. Extracts the binary number, which is the coded (and quantised) amplitude of the sample from which it was derived, from the frame.
3. Identifies the quantisation level which this number represents.
4. Generates a voltage proportional to this amplitude level.
5. Presents this voltage to the output V_{OUT} . The voltage appears at V_{OUT} for the duration of the frame under examination.
6. Message reconstruction can be achieved, albeit with some distortion, by low pass filtering.

6.6 Encoding

At the encoder the sample-and-hold operation (before encoding) is executed periodically. It produces a rectangular pulse form. Each pulse in the waveform is of *exactly* the same amplitude as the message *at the sampling instant*.

But *it is not possible* to recover a *distortion less* message from these samples. They are *flat top*, rather than *natural* samples.

Call this the sampling distortion.

At the encoder the amplitude of this waveform was then *quantised*. It is still a rectangular pulsed waveform, but the amplitude of each pulse will, in general, be in error by a small amount. Call this waveform $s(t)$.

6.7 Decoding

The voltage at V_{OUT} of the decoder is *identical with* $s(t)$ above. The decoder itself has introduced no distortion of the received signal.

But $s(t)$ is already an inexact version of the sample-and-hold operation at the encoder. This will give rise to *quantisation distortion* as well as the *sampling distortion* already mentioned.

You should read about these phenomena in the Text book.

7 The TIMS PCM Decoder Module

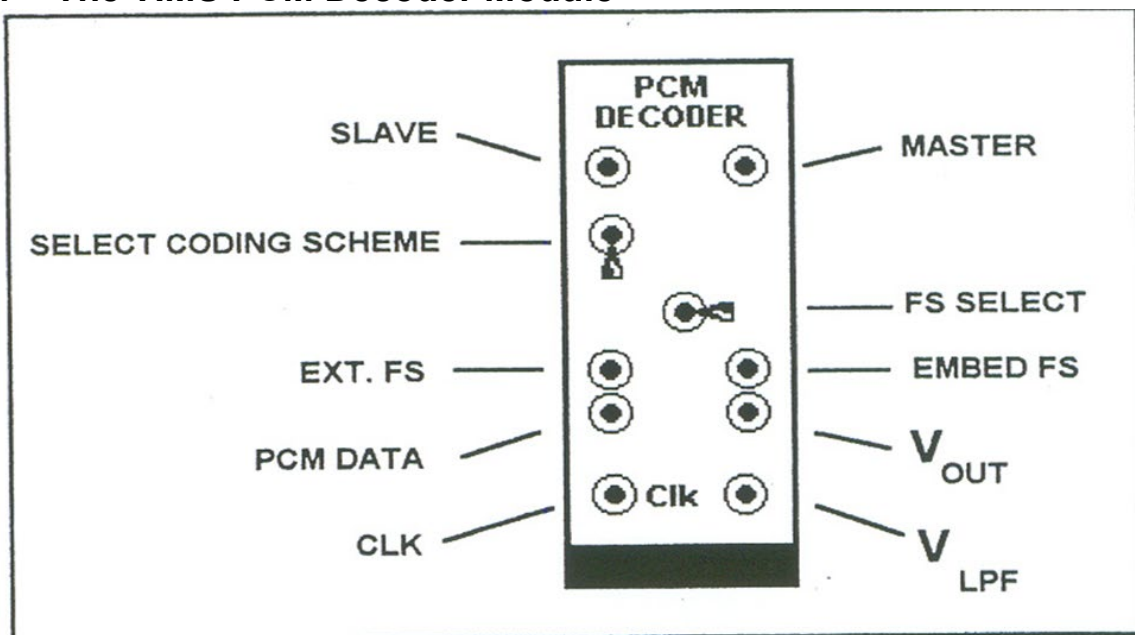


Figure 14: front panel layout of the PCM DECODER

A TIMS PCM DECODER module will be used for decoding.

The front panel of this module is shown in the Figure 9 above.

The connections are –

1. **SELECT CODING SCHEME:** a three-position toggle which selects the coding scheme used by the signal to be decoded
2. **FS SELECT:** a two-position toggle switch which selects the method of obtaining the frame synchronisation signal (FS) either external at (EXT.FS),

or derived internally from the embedded information in the received PCM itself (EMBED FS).

3. *EXT. FS*: connect an external frame synchronisation signal here if this method of frame synchronisation is to be used.
4. *EMBED FS*: if the frame synchronisation signal is derived internally from the embedded information, it is available for inspection at this output.
5. *PCM DATA*: the PCM signal to be decoded is connected here.
6. *VOUT*: the decoded PCM signal.
7. *CLK*: this is a TTL (red) input, and serves as the MASTER CLOCK for the module. Clock rate must be 10 kHz or less. For this experiment you will use the 8.333 kHz TTL signal from the MASTER SIGNALS module.

8 Experiment

8.1 The transmitter (encoder)

A suitable source of PCM signal will be generated using a PCM ENCODER module.

You should set it up before patching up the demodulator.

- (a). Use the 8.333 kHz TTL signal from the MASTER SIGNALS module for the CLK.
- (b). Select, with the front panel toggle switch, the 4-bit LINEAR coding scheme.
- (c). Synchronise the oscilloscope 'externally' to the frame synchronisation signal at FS. Set the sweep speed to 0.5 ms/div (say). This should show a few frames on the screen.
- (d). Connect CH 1 of the SCOPE SELECTOR to the PCM OUTPUT of the PCM ENCODER.
- (e). We would like to recognise the PCM DATA out signal. So choose a 'large' negative DC for the message (from the VARIABLE DC module). From previous work we know the corresponding code word is '0000', so only the embedded alternating '0' and '1' bits (for remote FS) in the LSB position should be seen. Confirm this. They should be 1920 ms apart. Confirm this both by measurement and calculation!
- (f). Vary the DC output and show the appearance of new patterns on CH 1. When finished, return the DC to its maximum negative value (control fully anti-clockwise).

The PCM signal is now ready for transmission.

It will be connected directly to a TIMS PCM DECODER module.

8.2 The receiver (decoder)

- (g). Use the front panel toggle switch to select the 4-bit LINEAR decoding scheme (to match that of the transmitter)
- (h). 'Steal' an 8.333 kHz TTL clock signal from the transmitter and connect it to the CLK input.
- (i). In the first instance 'steal' the frame synchronisation signal FS from the transmitter by connecting it to the frame synchronisation input FS of the receiver. At the same time ensure that the FS SELECT toggle switch on the receiver is set to EXT. FS.
- (j). Ensure both channels of the oscilloscope are set to accept DC; set their gains to 1 volt/div. With their inputs grounded set their traces in the centre of their respective halves of the screen. Remove the grounds.

- (k). Connect CH 2 to the sample-and-hold output of the PCM DECODER.

8.3 A DC message

You are now ready to check the overall transmission from transmitter input to decoder output. The message is a DC signal.

- (l). Connect the PCM DATA output signal from the transmitter to the PCM DATA input of the receiver.
- (m). Slowly vary the DC output from the VARIABLE DC module back and forth over its complete range. Observe the behaviour of the two traces. The input to the encoder moves continuously. The output from the decoder moves in discrete steps. These are the 16 amplitude quantising steps of the PCM ENCODER.

You are observing the source of quantising noise. The output can take up only one of 16 predetermined values.

- (n). Draw up a table relating input to output voltages.

You can now see the number of quantising levels at the transmitter, and their values.

- (o). Compare the quantising levels just measured with those determined in the experiment entitled PCM encoding.

8.4 A periodic message

It was not possible, when examining the PCM ENCODER earlier in the experiment to see the sample-and-hold waveform within the *encoder*. But you have just been looking at it (assuming perfect decoding) at the output of the *decoder*.

With a periodic message its appearance may be more familiar to you.

- (p). Change to a periodic message by connecting the SYNC MESSAGE of the PCM ENCODER, via a BUFFER AMPLIFIER, to its input V_{in} . An amplitude of 2 Vpp is suitable. Slow down the oscilloscope sweep speed to 1 ms/cm. Observe and record the signal at CH 2.

8.5 Message reconstruction

You can see, qualitatively, that the output is related to the input. The message could probably be recovered from this waveform. But it would be difficult to predict with what accuracy.

Low-pass filtering of the waveform at the output of the decoder will reconstruct the message, although theory shows that it will not be perfect. It will improve with the number of quantising levels.

Question 7. What amplitude characteristic is required for the reconstruction filter? If any distortion components are present they would most likely include harmonics of the message. If these are to be measurable (visible on the oscilloscope, in the present case), then they must not be removed by the filter and so give a false indication of performance, so we could look for harmonics in the output of the filter.

- (q). Change the oscilloscope to the FFT mode and look for 2nd. and 3rd. order harmonics. Is there distortion ?

Questions

- Question 8. *How would you arrive at a specification for the reconstruction filter used in this experiment?*
- Question 9. *Two sources of distortion of the reconstructed message have been identified - they were called sampling distortion and quantising distortion. (a) assuming a sample-and-hold type sampler, what can be done about minimising sampling distortion? (b) what can be done about minimising quantising distortion?*
- Question 10. *Quantising distortion decreases with the number of quantising levels available. There is usually a price to be paid for such an option. What would this be? Was that apparent in the present experiment? Explain.*

LAB 5 AMPLITUDE MODULATION USING SIMULINK

1 Objective

To be familiar with Simulink (Matlab) using amplitude modulation as an example.
[http://ocw.usu.edu/Electrical_and_Computer_Engineering/Communication_Systems_I_1/lab1.html]

2 Theory

Amplitude modulation (AM) is a modulation technique used in electronic communication, most commonly for transmitting information via a radio carrier wave. In amplitude modulation, the amplitude (signal strength) of the carrier wave is varied in proportion to the waveform being transmitted. That waveform may, for instance, correspond to the sounds to be reproduced by a loudspeaker, or the light intensity of television pixels. This technique contrasts with frequency modulation, in which the frequency of the carrier signal is varied, and phase modulation, in which its phase is varied.

3 Equipment

1. Desktop PC with Matlab installed.
2. Lecture notes related to amplitude modulation

4 Method and Questions

Be familiar with Matlab and Simulink by watching this online video (URL: <https://www.youtube.com/watch?v=TfXBw51xKr0>) or read this document (URL: http://faculty.washington.edu/lum/website_professional/matlab/tutorials/Simulink_Tutorial/simulink_tutorial.pdf).

4.1 Start Simulink Programme

Start up Matlab by double-clicking the shortcut on the desktop.

Start up Simulink by typing

`>> Simulink`

at the Matlab prompt. You should see the Simulink Block Library windows as shown below.

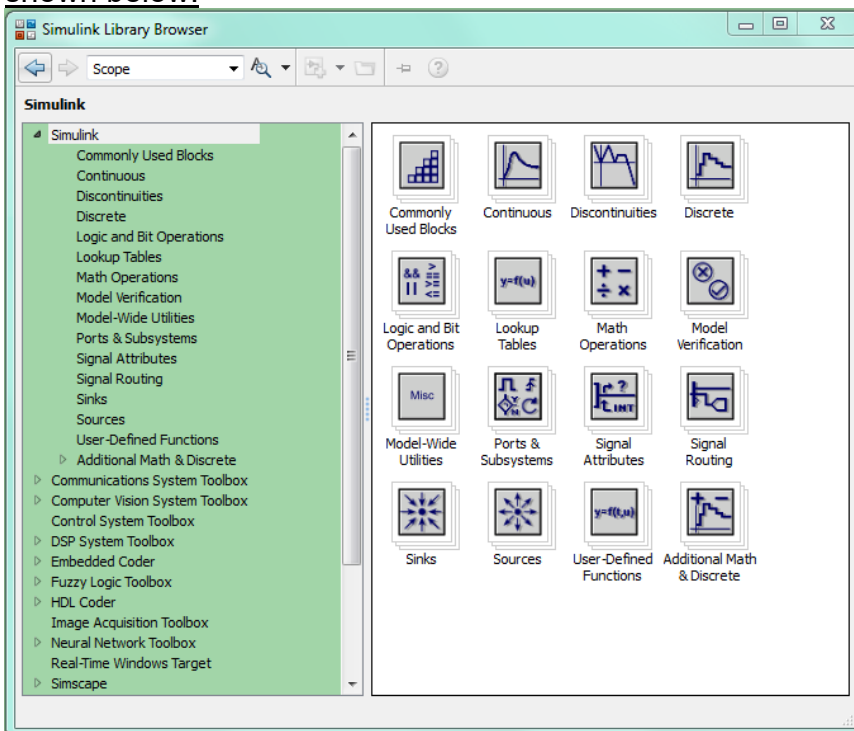


Figure 15. Simulink Block Library

4.2 Create Simulink Model for AM-DSB/SC

From the file menu, select New -> Simulink Model.

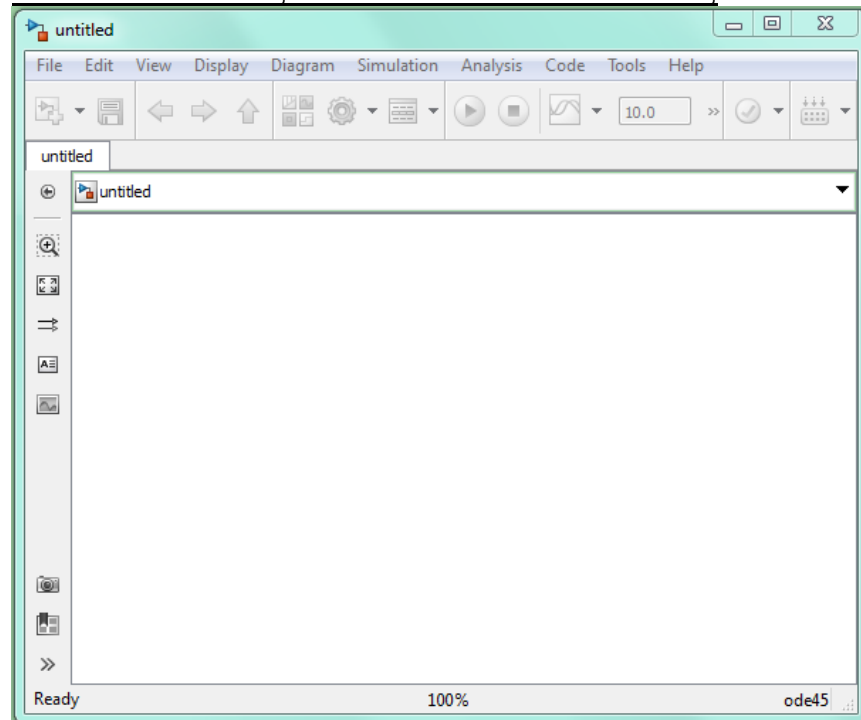


Figure 16. New Simulink Model

We will start by building and testing an AM-DSB/SC modulator. Click on the Sources block in the Simulink Block Library window. The Sources Block Library should open as shown below.

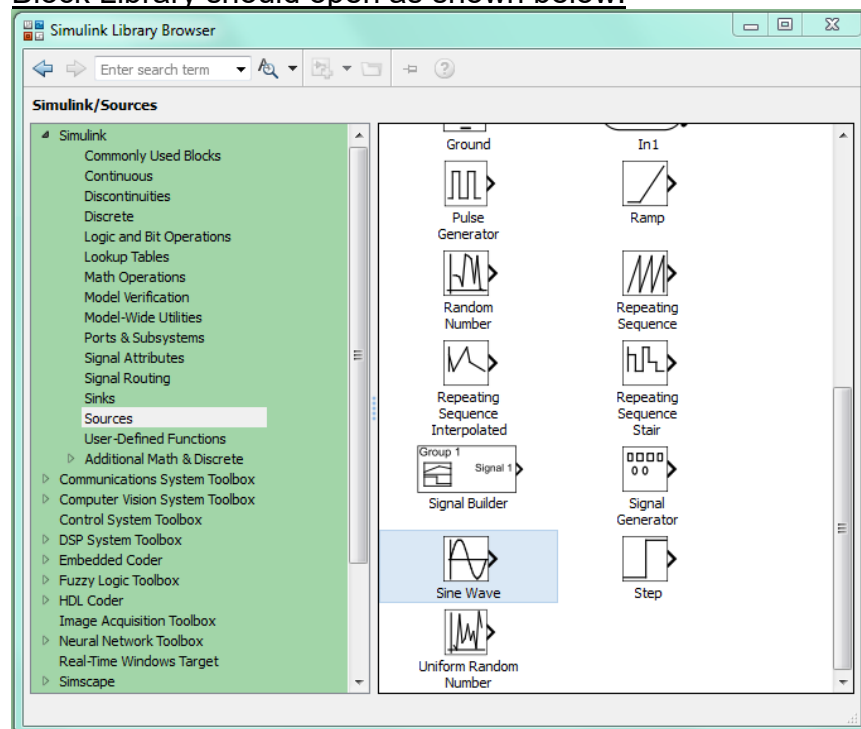


Figure 17. Simulink Source Library

Drag and drop a **Sine Wave** block from the Sources Library to the Model window. Click on **DSP System Toolbox**, then **Sources**, you will see this

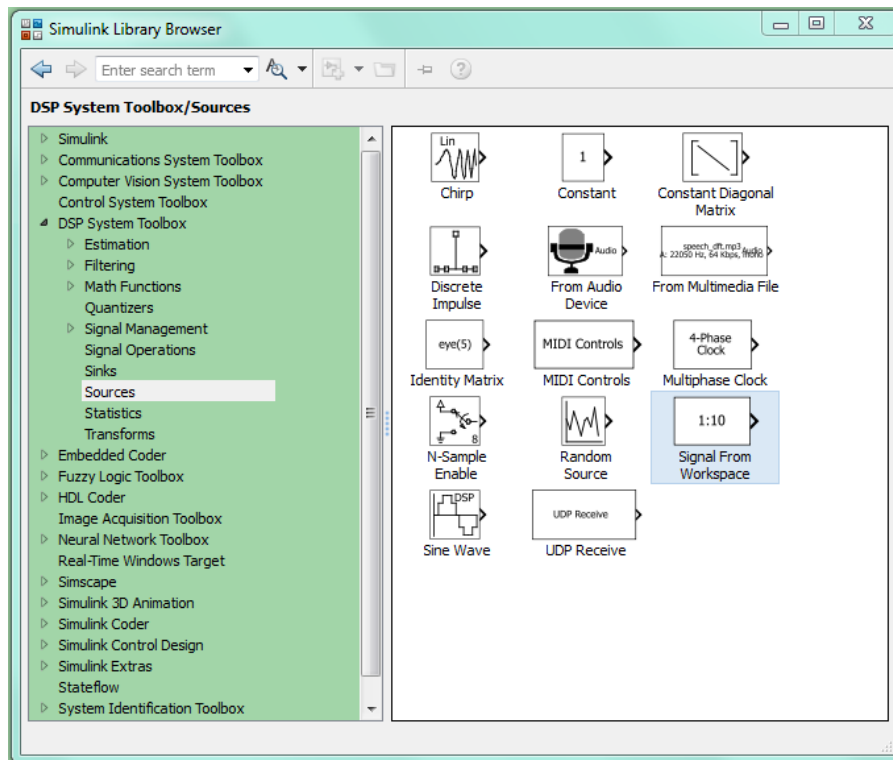


Figure 18. DSP Source Library

Drag and drop the **Signal From Workspace** block into the Model window.
Click on the Simulink Block Library, then Math Operations.

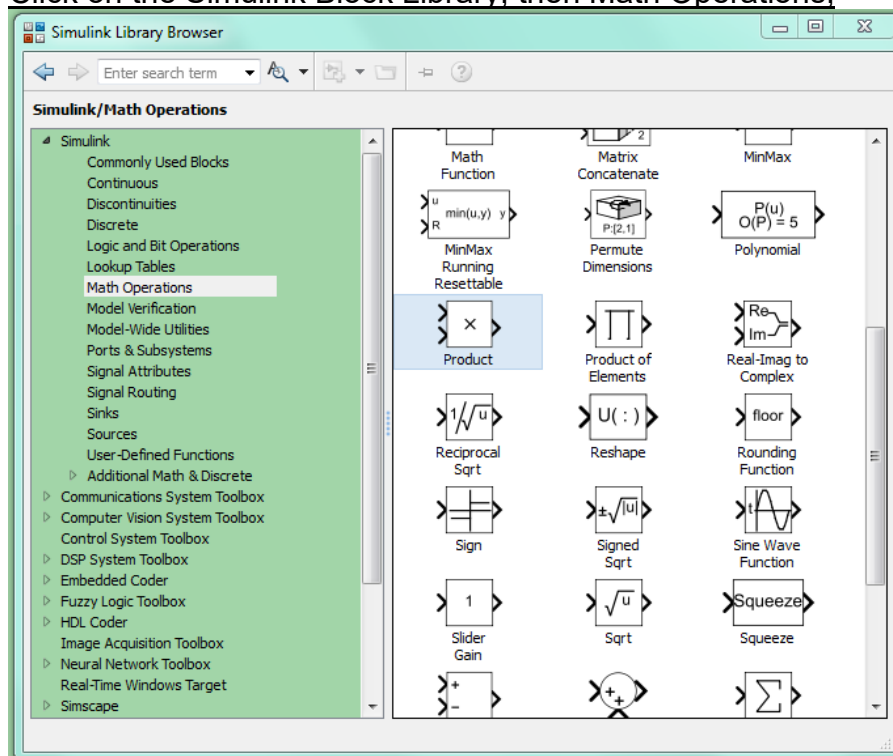


Figure 19. Simulink Math Operations Library

Drag and drop the Product block onto the Model window.
Click on the Sinks block in the Simulink Block Library window. The Sinks Block Library window should open as shown below.

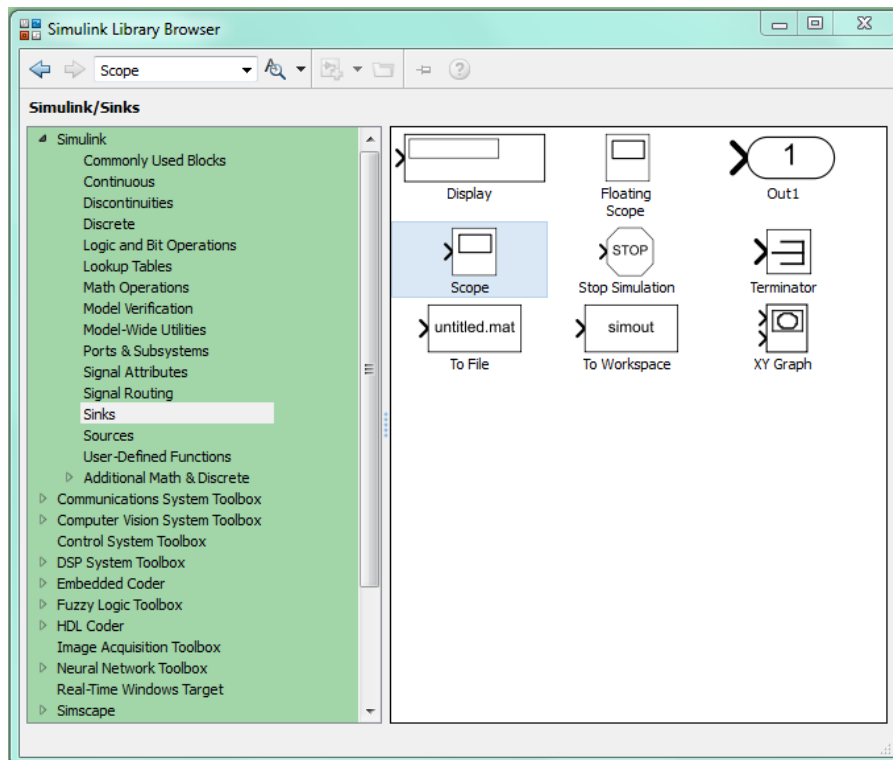


Figure 20. Simulink Sink Library

Connect the blocks with signal flow lines. The Model window should now look like

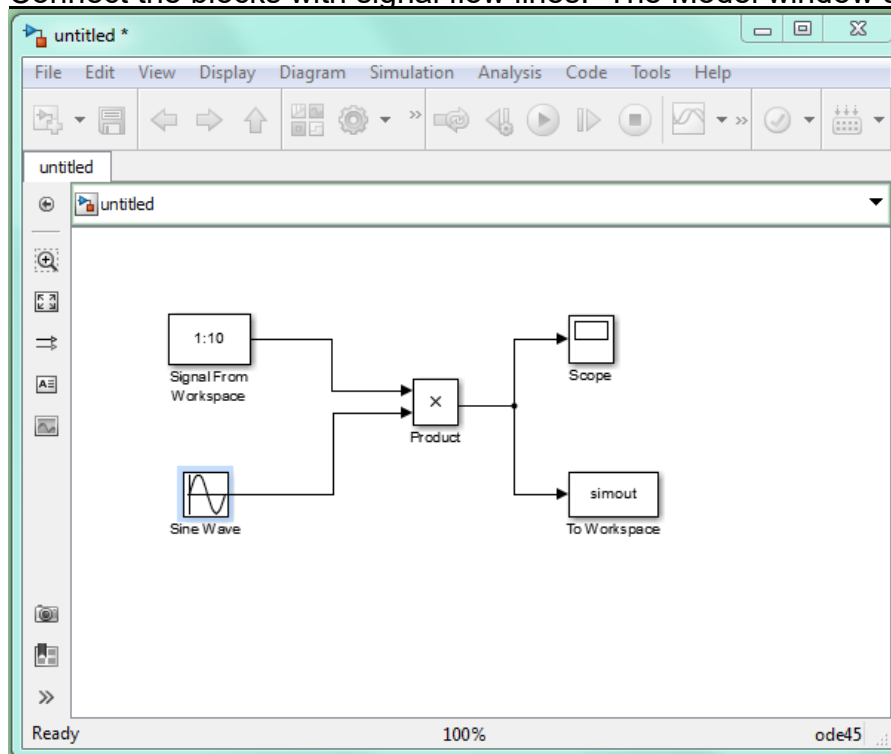


Figure 21. Built Simulink Model

4.3 Setting up the Simulation

Open the Sine Wave block by double clicking on it. Set the parameters as follows.

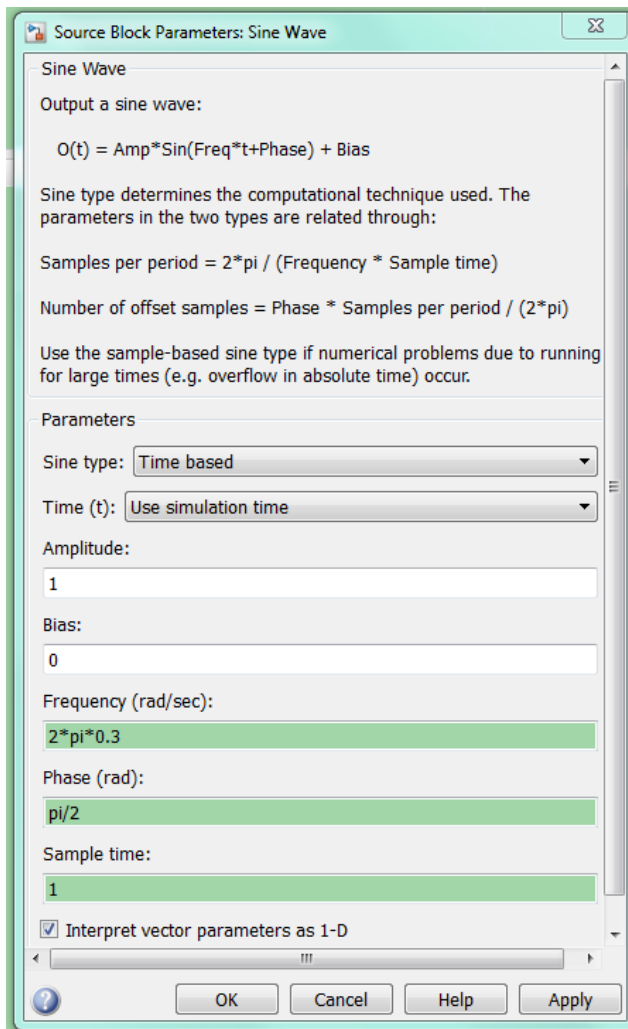


Figure 22. Sinusoidal Source Property Window

Click OK when you are done to set the parameters of the Sine Wave block.
In the Matlab Command Window define the signal bumps by typing the following.

```
>> bump = sqrt(1250^2-([0:2499]-1250).^2) / 250;
>> bumps = [bump bump];
>> plot(bumps)
```

The resulting plot should look something like this. Do you recognize the McDonelld's "Golden Arches"?

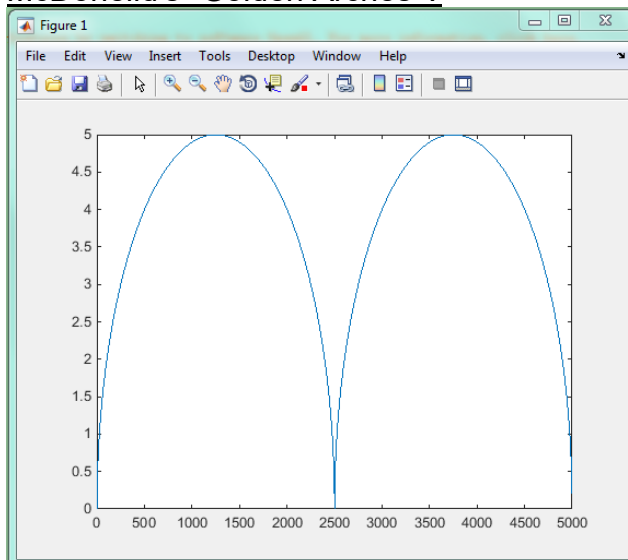


Figure 23. Message Information

This signal is 5000 samples long and will be used as the modulating signal for our AM-DSB/SC modulator. To let Simulink know that we want this for our source signal, double click the From Workspace block in the Model window and set the Parameters as shown below.

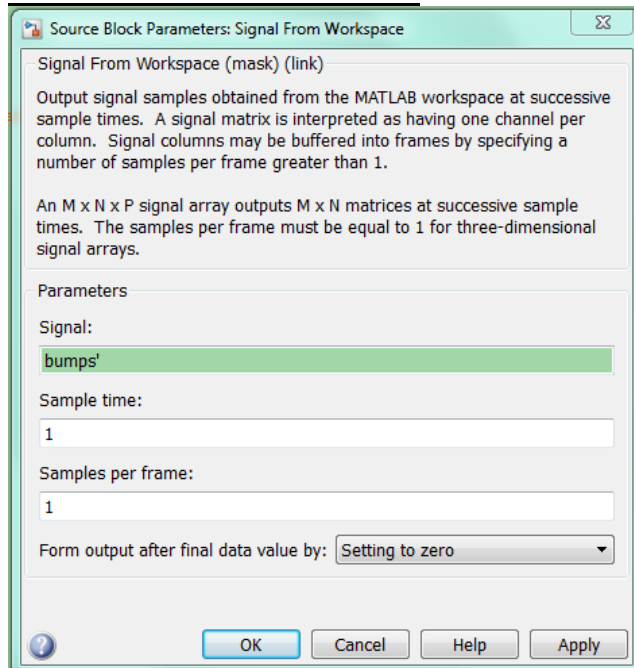


Figure 24. Signal from Workspace Property Window

Double click the To Workspace block in the Model window and set the Parameters as shown below.

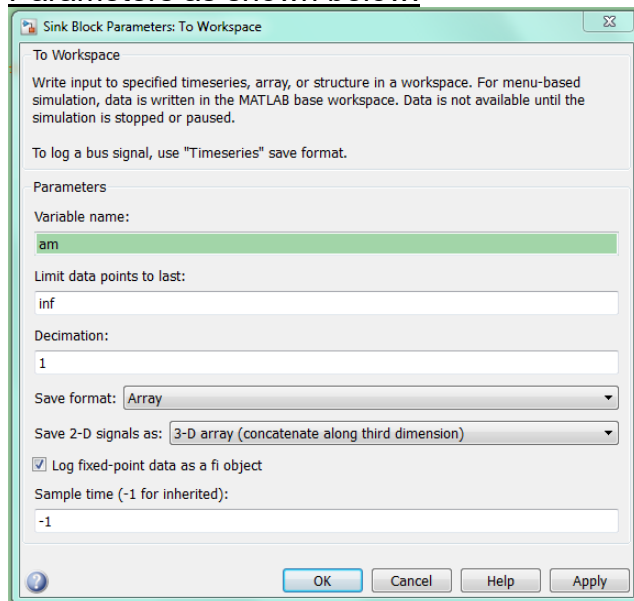


Figure 25. Output to Workspace Property Window

Click OK to set the values.

4.4 Save Simulink Model and Configure Simulation Parameters

Save your model by selecting Save as... in the File menu in the Model window. Call this model **lab5.slx**.

Bring up the simulation parameters dialog from the **Simulation** menu in the Model window, select **Model Configuration Parameter** (Ctrl+E).

Under Simulation time, change the Stop time to **5000**.

Under Solver options, change the Type to Fixed-step.

The Configuration Parameters window should look something like this.

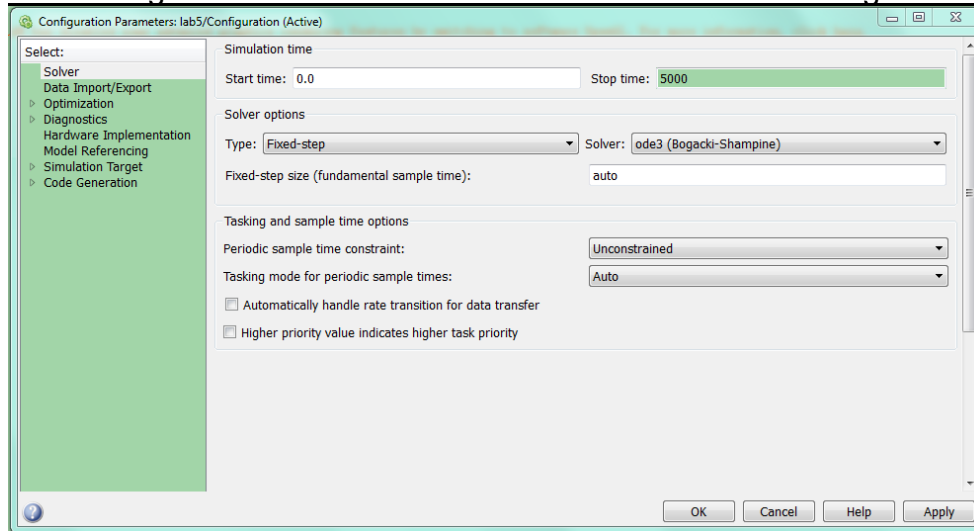


Figure 26. Simulation Configuration Parameters Window

Set the values by clicking OK. Next select **Run** (Ctrl+T) from the Simulation menu in the Model window.

4.5 Simulation Results

Open the Scope display by double clicking on the scope. Then you should see something like this.

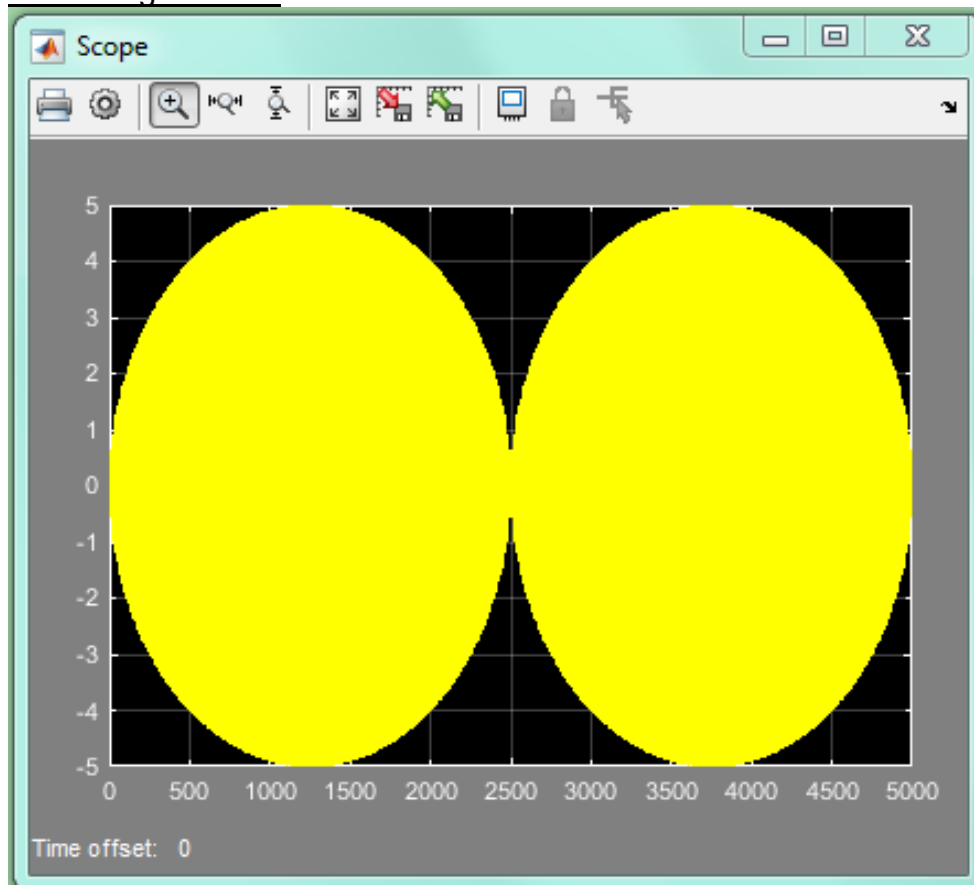


Figure 27. AM-DSB/SC Output Waveform

You can zoom in to verify that this is an amplitude modulated wave corresponding to the modulating signal given above. By entering the following in the Matlab Command Window, you can see that the original modulating signal has been moved from baseband to a carrier frequency at 0.3.

```
>> NFFT=2^16;
>> f=[0:NFFT-1]/NFFT;
>> AM=abs(fft(out.am(1,:).*blackman(length(out.am)).',NFFT));
>> plot(f,20*log10(AM));
>> axis([0 0.5 -20 80]);
```

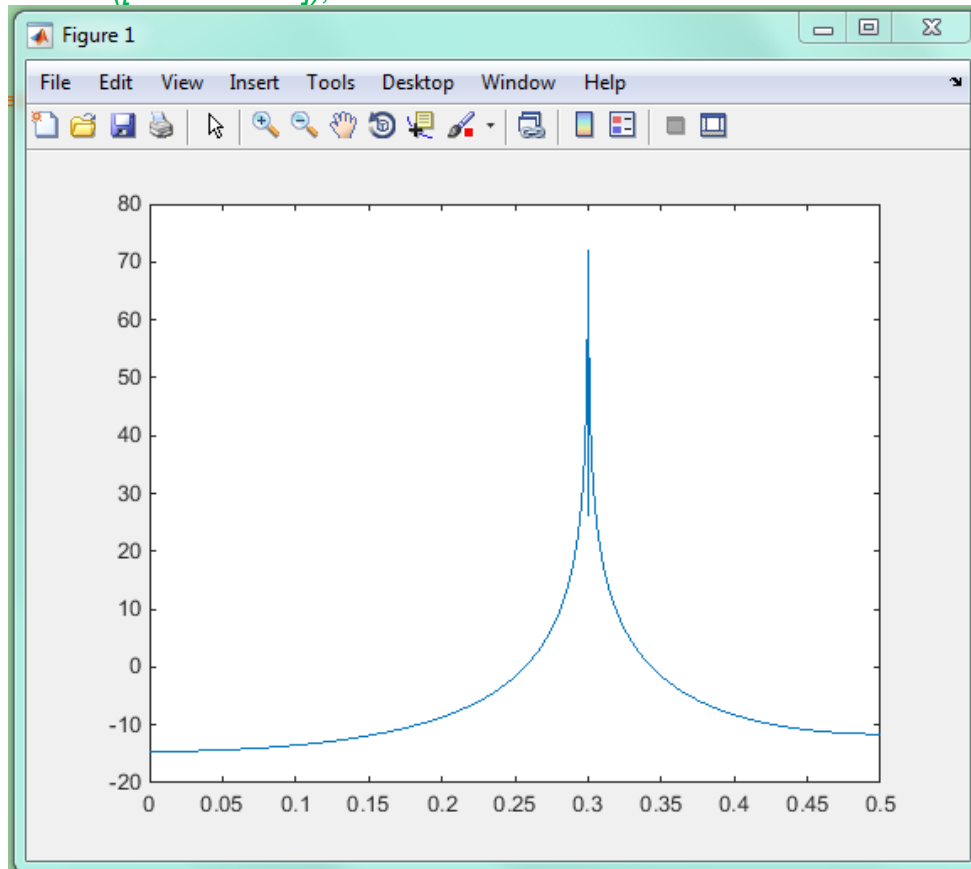


Figure 28. Output Spectrum

5 Questions

Question 1. Write a Matlab function that will display the spectrum of a signal.

Here's some code that you may want to try:

```
function sigspec(signin,flag)
if(nargin==0)
    disp('USAGE: sigspec(signin,flag)');
    disp(' The input signals should be in the columns of signin. ');
    disp(' flag=0 produces a linear magnitude plot (default)');
    disp(' flag=1 produces a log magnitude plot (dB)');
    return;
end

if(nargin==1)
    flag = 0;
end
[len,numsigns]=size(signin);
if(len<numsigns)
    signin = signin.'; % Force the signals into the columns of signin
    tmp = len;
    len = numsigns;
    numsigns = tmp;
end
```

```

fftsize = 2^(ceil(log2(len))+2); % Zero-padded FFT size
f=[0:fftsize-1].'/fftsize - 0.5; % Normalized frequency
signin = repmat(hamming(len),1,numsigns).*signin; % Window to reduce spec.
leakage
siginspec = abs(fftshift(fft(signin,fftsize),1)); % Compute spectrum
if(flag)
    plot(f,10*log10(siginspec));
else
    plot(f,siginspec);
end
xlabel('Normalized Frequency');
ylabel('Magnitude (dB)');

```

Question 2. *In the simulink model created above, there are three signals: (1) the modulating signal, (2) the carrier signal, (3) the modulated signal. The modulating signal and the modulated signal should already exist in the Matlab workspace. Add another To Workspace block to the model and connect it to the carrier signal. Run the simulation. Then all three signals should exist in the Matlab workspace. Use the sigspec.m function that you wrote previously to plot the spectra of the three signals. Plot all three signals in the same figure window but in different axes. Use the subplot function in Matlab to accomplish this. If you have never used subplot, at the Matlab prompt, type*

>> help subplot

Write an explanation of the three plots. Comment on the scaling of the modulated signal relative to the modulating signal. The scaling should be one half in the linear scale and -3dB on the log scale. To get an accurate measurement, you may need to zoom in to read off the correct value of the peak. A log plot would look something like this.

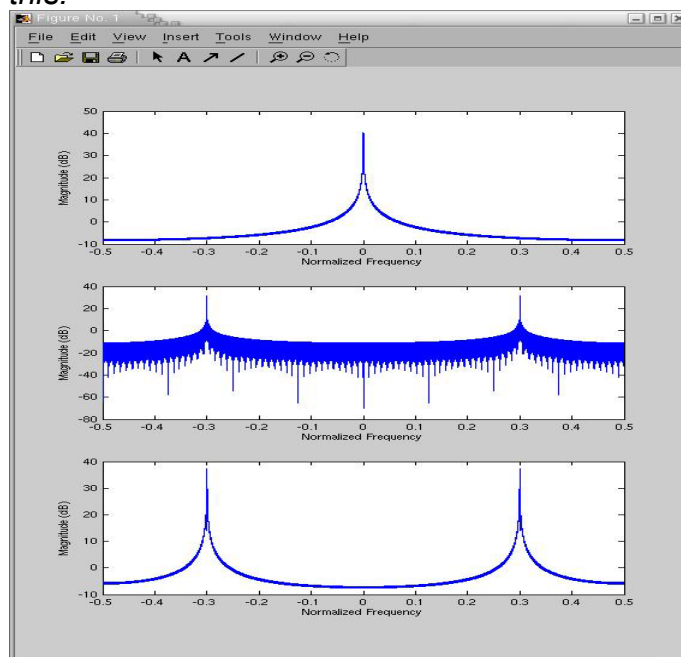


Figure 29. Output Spectra

LAB 6 HARMONIC ANALYSIS

1 Objective

To determine the various component frequencies present in a rectangular waveform.

2 Theory

A sine wave can be described as

$$Y = A \sin \omega t \quad (3)$$

That gives a sine wave has an amplitude (A) and a frequency (f) derived from

$$\omega = 2\pi f \quad (4)$$

This waveform can be viewed on an oscilloscope which displays the waveform in the time domain (amplitude versus time). However, the same waveform can be displayed in the frequency domain (amplitude versus frequency). The time domain can be converted to the frequency domain and vice versa.

Any complex waveform i.e., one that is not sinusoidal (for example - square, triangle, speech waveforms etc.) can be analysed in the frequency domain. This is commonly known as the harmonic analysis.

Consider a periodic rectangular pulse, which when described as a function of time has the form shown below.

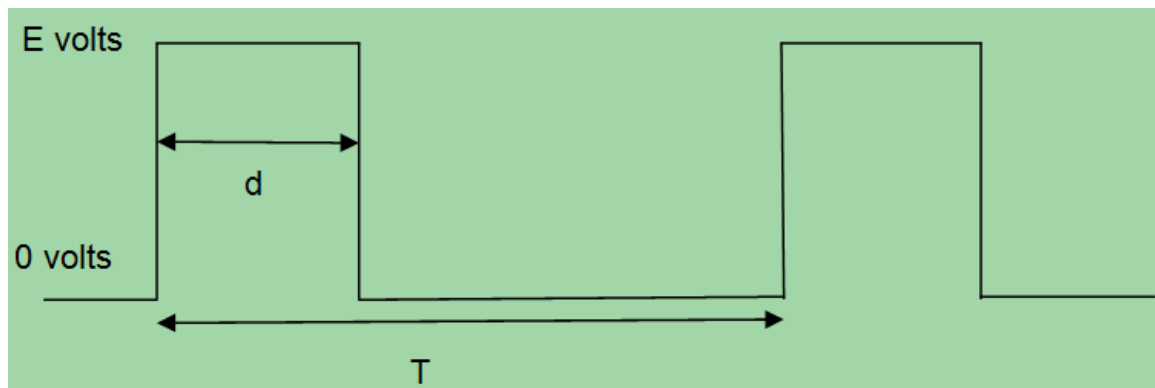


Figure 30: Periodic rectangular pulse

The pulse lasts for a time d and is repeated after a time T so that the frequency of repetition, or pulse repetition frequency (PRF) is defined as $1/T$. The duty cycle or duty ratio is defined as d/T .

The complex waveform can be broken down into a number of harmonics each having a fixed amplitude and frequency. The amplitude may be expressed as a voltage (or current) or as a ratio (in dB) with respect to the fundamental.

The frequency of the harmonic may be expressed directly or expressed as a harmonic number with respect to the fundamental.

For example - the fundamental (repetition frequency) may be 15 kHz and signals are measured at 30 kHz - 2nd harmonic, 45 kHz - 3rd harmonic and 60 kHz - 4th

harmonic, and so on. Thus the harmonic numbers are 1 (fundamental) 2, 3, 4, .., etc..

The square wave and rectangular wave can be expressed mathematically as:

$$y = a_0 + \sum_{n=1}^{\infty} a_n \sin(n\omega t) + \sum_{n=1}^{\infty} b_n \cos(n\omega t) \quad (5)$$

The amplitudes of the harmonics can be determined by expression:

Amplitude of nth harmonic

$$a_n = \frac{2Ed}{T} \text{sinc}\left(\frac{n\pi d}{T}\right) = \frac{2E}{n\pi} \sin\left(\frac{n\pi d}{T}\right), \quad b_n = 0 \quad (6)$$

This expression is best worked out by using a spread-sheet program such as Excel or by using Matlab.

Other complex waveforms may be expressed by similar equations.

3 Method

- (a). A pulses with the following duty ratios (d/T) are to be investigated. duty ratios (d/T) of:- 1:2, 1:3, 2:3, 1:5, 1:10, and 1:20.
- (b). Using EXCEL or Matlab, calculate the amplitude of the harmonics up to a value of a least $n = 3T$ ie. for a duty ratio of 1:5, calculate the amplitude of all the harmonics up to and including the 15th. harmonic.
- (c). Plot a graph in histogram (bar chart) form of Amplitude (as given by the program) versus harmonic number. Calculate the DC component and plot this information on the graph. The graphs must have a title, and the scales labeled.

4 Questions

- Question 1. *Derive the expression for the DC value of the general pulse waveform shown on the first page of this experiment.*
- Question 2. *In Question 1 above, what would be the positive peak value if the DC value was zero?*
- Question 3. *Derive an expression, involving d, T, n, for the harmonic number that has zero amplitude.*
- Question 4. *What is the effect on the frequency domain graph if the duty cycle decreases but the frequency remains constants?*
- Question 5. *What is a unit impulse?*
- Question 6. *What frequency domain graph would a unit impulse give?*

Appendix A RECEIVER AND TRANSMITTER BLOCK DIAGRAMS

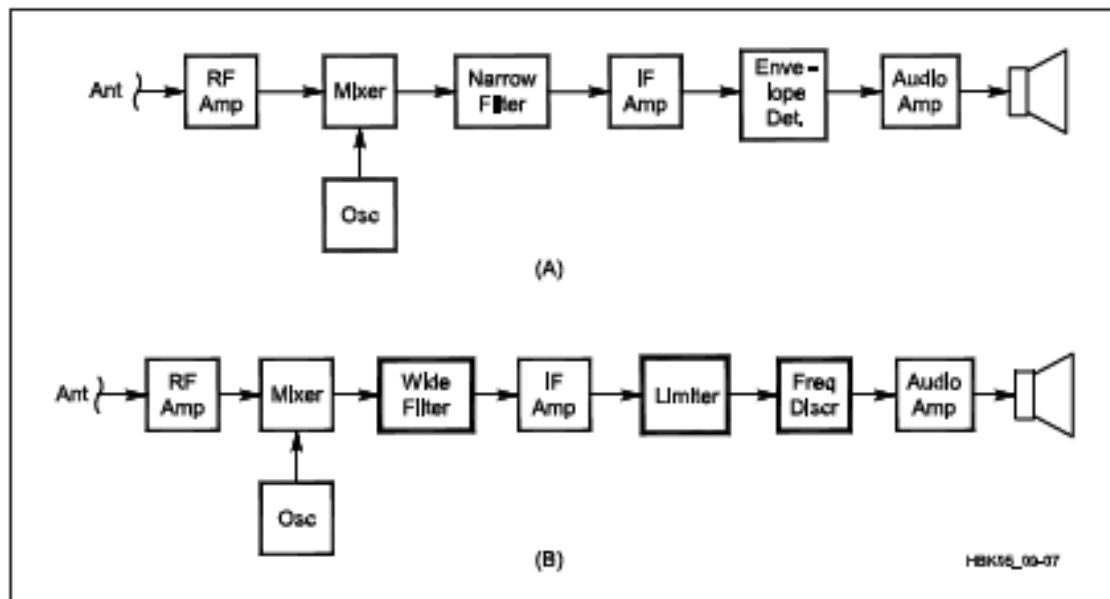


Figure 31: (A) a block diagram of an AM receiver. (B) an FM receiver.

Note: The dark borders outline the sections that are different in the FM receiver.

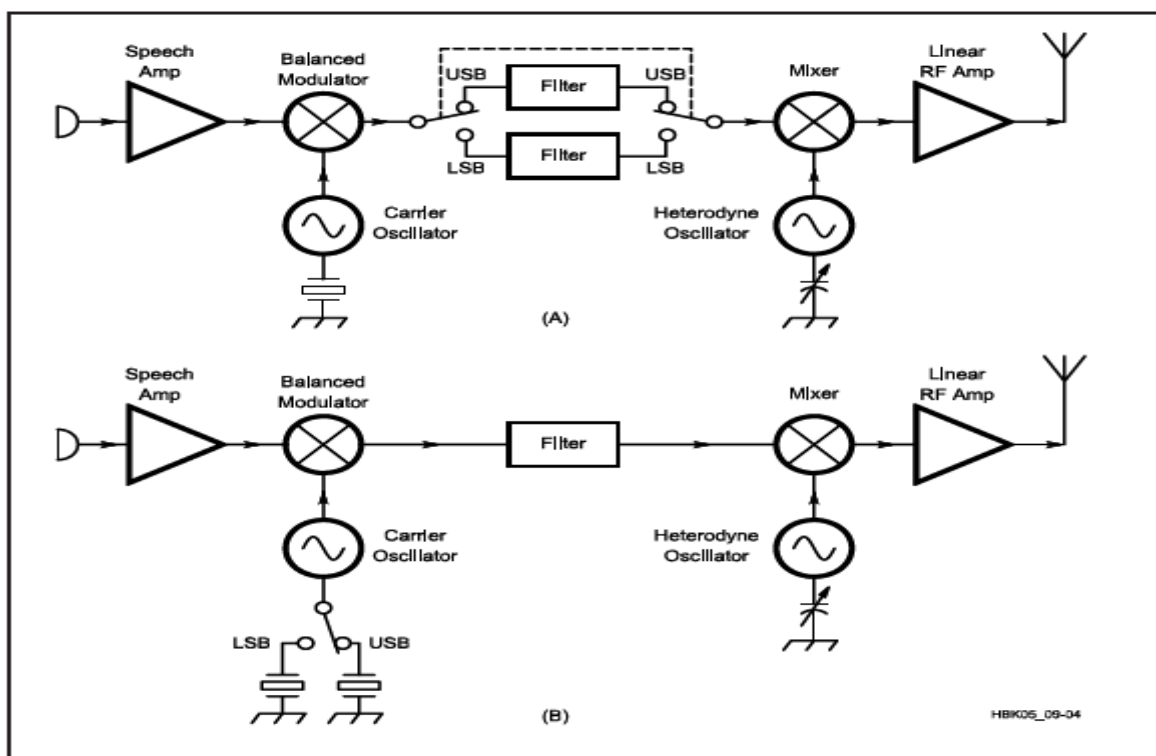


Figure 32: Block diagrams of the filter method of SSB generation.

Note: They differ in the manner that the upper and lower sideband are selected.

The following figures show typical circuits of standard communication blocks. While the circuits use discrete components, modern ICs (in many cases) use similar circuit. Here in this section, we are looking at principles. There are many alternatives to these circuits.

1 Modulator

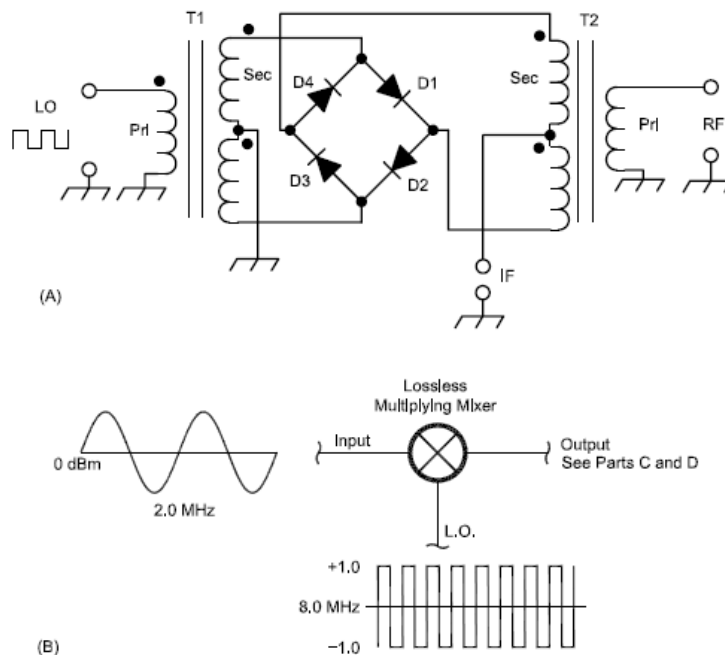


Figure 33: AM modulator. (A) a general-purpose diode reversing-switch Mixer (Ring bridge double balanced modulator). This mixer can use a square-wave LO and a sine-wave input signal. (B) the input waveforms.

The output waveform is the typical DBM waveform. This waveform contains many unwanted products of modulation that are removed by the following band-pass filter. The circuit is simple, cheap, performs well into the UHF range and is available in IC form.

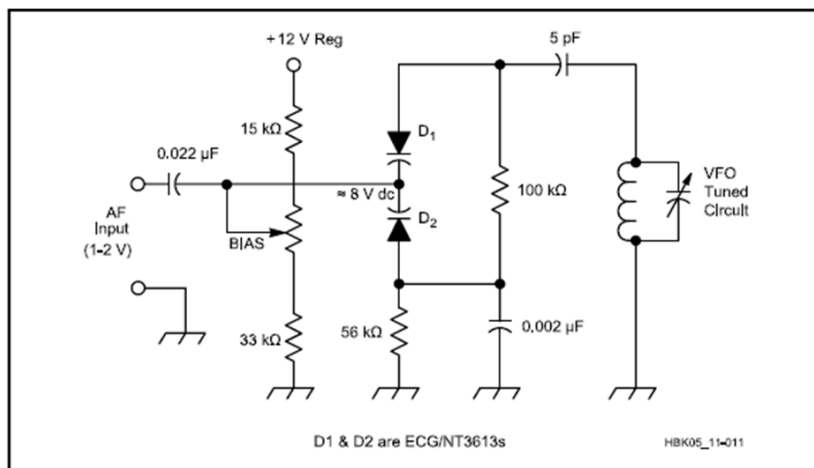


Figure 34: One or more tuning diodes can serve as the variable reactance in a reactance modulator. This HF reactance modulator circuit uses two diodes in series to ensure that the tuned circuit's RF-voltage swing cannot bias the diodes into conduction. D1 and D2 are "30-volt" tuning diodes that exhibit a capacitance of 22 pF at a bias voltage of 4. The BIAS control sets the point on the diode's voltage versus-capacitance characteristic around which the modulating waveform swings.

2 Demodulator

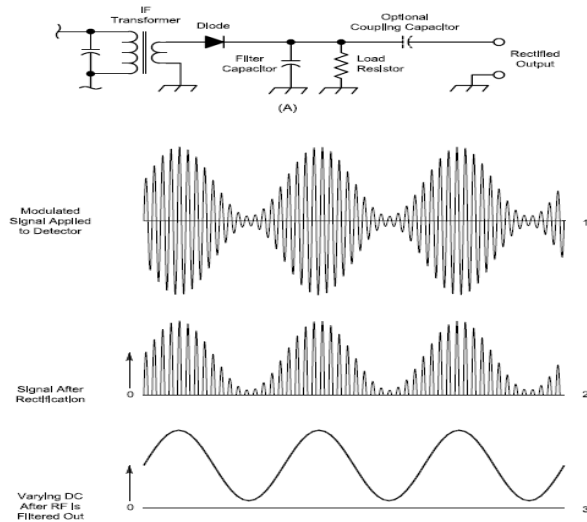


Figure 35: Radio's simplest demodulator, the diode rectifier (A), demodulates an AM signal by multiplying its carrier and sidebands to produce frequency sums and differences, two of which sum into a replica of the original modulation.

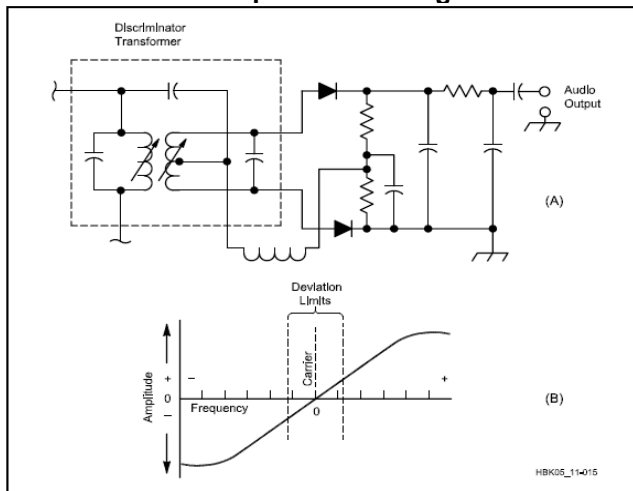


Figure 36: A discriminator (A) converts an angle-modulated signal's deviation into an amplitude variation (B) and envelope-detects the resulting AM signal. For undistorted demodulation, the discriminator's amplitude-versus-frequency characteristic must be linear across the input signal's deviation. Again, modern FM detectors are based on ICs.

3 Oscillator

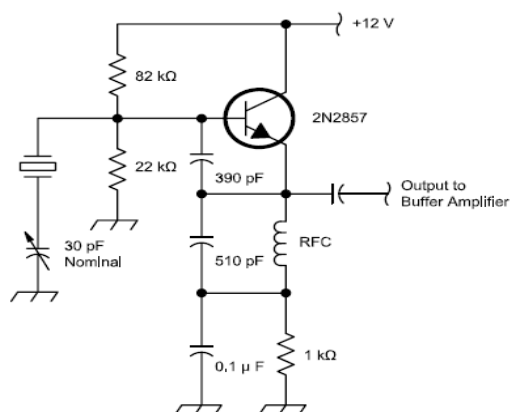


Figure 37: The circuit is of a crystal operating in its series mode in a series-tuned Colpitts circuit. Because it does not include an LC tank to prevent operation on unwanted modes, this circuit is intended for fundamental mode operation only and relies on that mode being the most active. Variable frequency oscillators (VFO) have a series tuned circuit in place of the crystal.

4 Filter

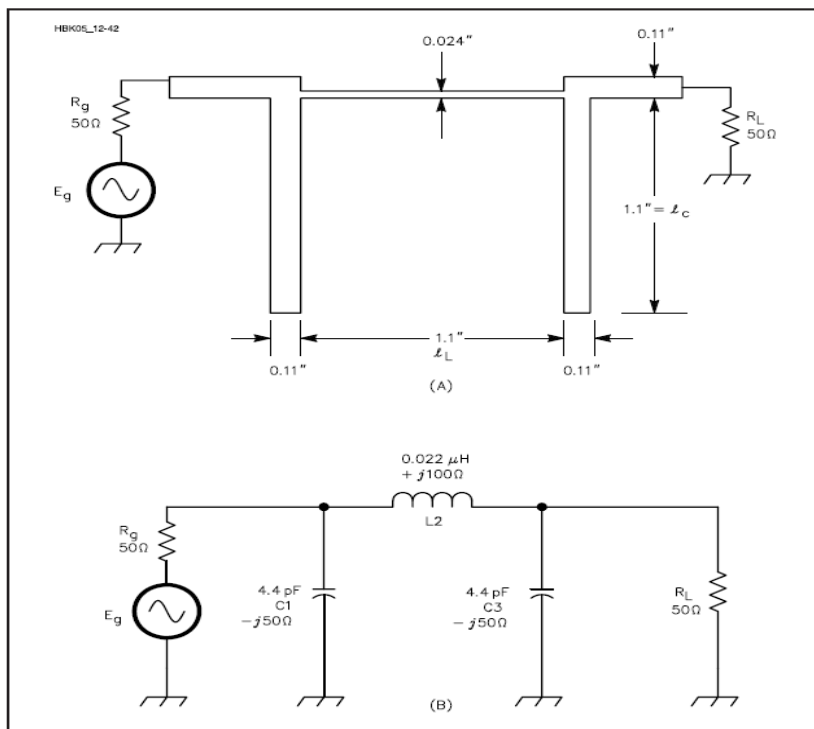


Figure 38: (A) microstrip 3-pole emulated-Butterworth low-pass filter with a cutoff frequency of 720 MHz. (A) shows a microstrip version built with G-10 fiberglass board ($\epsilon = 4.5$, $h = 0.062$ inches). (B) shows the Lumped LC version of the same filter. To construct this filter with lumped elements very small values of L and C must be used and stray capacitance and inductance must be reduced to a tiny fraction of the component values.

5 Small-signal amplifier.

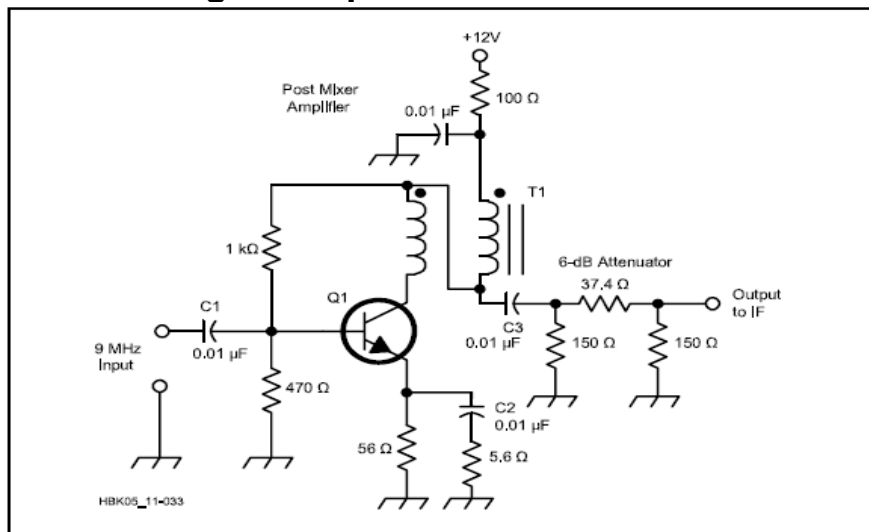


Figure 39: This amplifier's gain, including the 6-dB loss of the attenuator pad, is about 16 dB; its noise figure, 4 to 5 dB; its output intercept, 30 dBm. The 6-dB attenuator is essential if a crystal filter follows the amplifier; the pad isolates the amplifier from the filter's highly reactive input impedance.

Appendix B WAVEFORM SPECTRA

1 Introduction

Signals are characterized by an amplitude/time variation of some physical quantity; for example, the ear detects the air pressure/time variation in a sound wave. When transmitted through a telecommunications channel, at some point the signal is converted to analogous voltage- and current-time variations, which for brevity will be referred to as the *signal waveform*. A fact of fundamental importance in communications is that the time waveform of a signal can be represented by a series of sine and/or cosine waves.

Such a representation is termed the *spectrum* of the signal. The reason the sine (or cosine) wave is chosen as the basic waveform type is because the response of a channel to this type of wave is easily determined mathematically and by measurement, and the results can be extended to the wave represented by the series of sine or cosine waves.

A sinusoidal waveform described by Equation (7) is shown in Figure 41(a).

$$v(t) = V_{\max} \sin(2\pi ft) \quad (7)$$

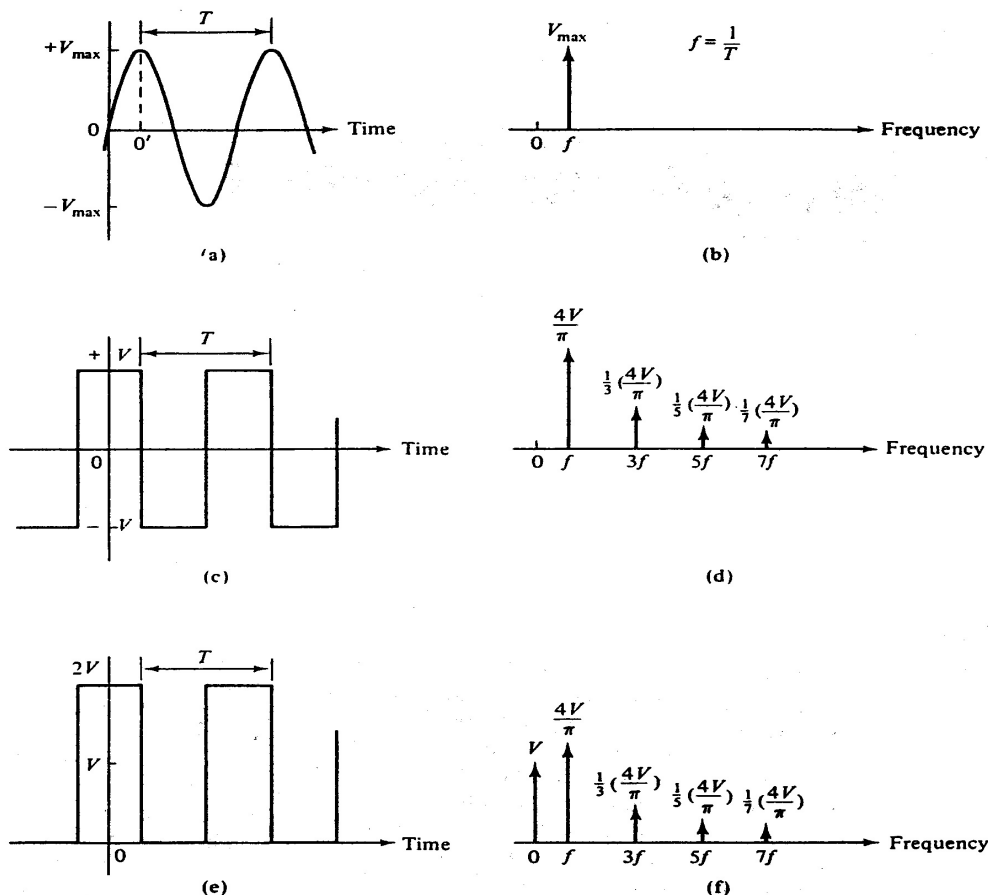


Figure 41: (a) A sinusoidal voltage waveform; (b) spectrum of a sinusoidal waveform; (c) a square voltage waveform; (d) magnitude spectrum of a square wave; (e) a square wave with a positive dc component; (f) magnitude / frequency spectrum for (e).

Alternatively, if the zero-time origin is started at $0'$ instead of 0 (as shown in Figure 41(a)), the wave can be described by

$$v(t) = V_{\max} \cos(2\pi ft) \quad (8)$$

Since the choice of time origin is arbitrary, the choice of representing the wave by a sine or cosine function will usually be unimportant. Again, it should be noted that Equation (7) can be expressed as

$$v(t) = V_{\max} \cos(2\pi ft - \pi/2) \quad (9)$$

The spectrum of the sinusoidal wave is simply a straight line of height $V_{\max}$ positioned at f on the frequency axis, as shown in Figure 41(b). The amplitude spectrum takes no account of whether the wave is represented by sine or cosine. A separate phase angle/frequency graph may be shown if this information is required.

2 Complex Repetitive Waves

Any waveform, other than the sine or cosine wave, which repeats itself at regular intervals is termed a *complex repetitive wave*. The period T over which the waveform repeats is termed the *periodic time*. The spectrum for any complex repetitive wave can be found by means of a mathematical method known as *Fourier analysis*, but only the results of Fourier analysis for some common types of repetitive waveforms will be used here.

The rectangular wave shown in Figure 41(c) can be represented by the Fourier series

$$v(t) = \frac{4V}{\pi} \left[\cos(\omega t) - \frac{1}{3} \cos(3\omega t) + \frac{1}{5} \cos(5\omega t) - \frac{1}{7} \cos(7\omega t) + \dots \right] \quad (10)$$

Note that the symmetry of the square wave about the axes is similar to a cosine wave, and as a result the series of Eq. (10) contains only cosine terms.

The series has an infinite number of terms, but it can be seen that the amplitude of each term decreases as $1/n$. It will also be seen that the series contains only odd harmonics (i.e., components at frequencies f , $3f$, $5f$, etc.). The spectrum for the rectangular wave is shown in Figure 41(d).

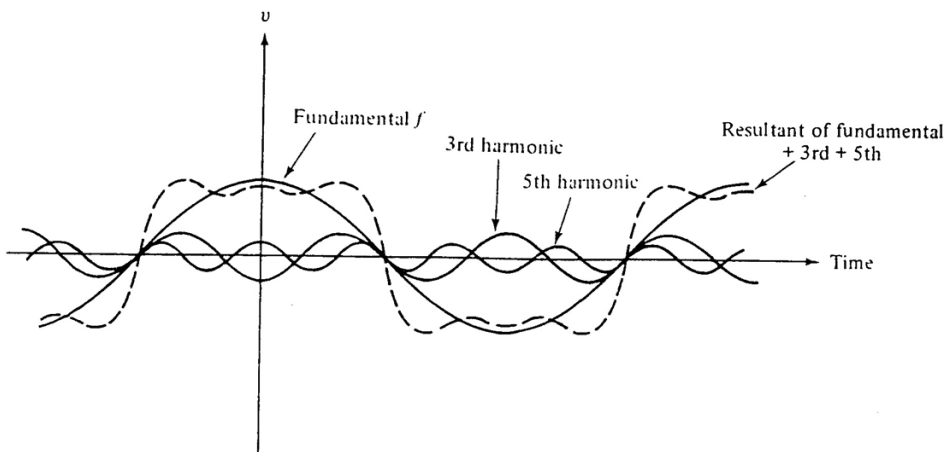


Figure 42: An illustration of how the three sinusoidal components add to approximate a square wave.

It must be clearly understood that the spectrum is more than just an alternative mathematical way of representing the wave. The component cosine waves, for

example, in the series for the rectangular wave, are as physically real as the original time waveform, and could be tuned out by means of frequency-selective filters. The first three components of the rectangular wave spectrum are shown in Figure 27. Adding these produces the resultant waveform shown dashed, and it will be seen that this approaches the rectangular wave shape.

By shifting the rectangular wave up on the voltage axis (Figure 41(e)), a zero-frequency component is added to the spectrum (Figure 41(f)). This zero-frequency component is just the average, or DC value of the waveform, and is the value that would be read on an average-reading voltmeter, such as a moving-coil meter. *Any waveform that is not symmetrical in area about the time axis will have a zero-frequency component.*

The series for the square wave of Figure 41(e) is

$$v(t) = V + \frac{4V}{\pi} \left[\cos(\omega t) - \frac{1}{3} \cos(3\omega t) + \frac{1}{5} \cos(5\omega t) - \frac{1}{7} \cos(7\omega t) + \dots \right] \quad (11)$$

thus showing the zero-frequency or DC component to be V .

The train of rectangular pulses in Figure 28(a) has the spectrum shown in Figure 28(b). This has a zero-frequency component and contains both odd and even harmonics (i.e., components at frequencies f , $2f$, $3f$, $4f$, etc. The amplitude of the spectrum components depends on the ratio τ/T of "on" time to periodic time, the series being given by

$$v = \frac{V\tau}{T} + \frac{2V\tau}{T} \left(\frac{\sin x}{x} \cos \omega t + \frac{\sin 2x}{2x} \cos 2\omega t + \frac{\sin 3x}{3x} \cos 3\omega t + \dots \right) \quad (12)$$

where $x = \pi\tau/T$. The series is written this way because the function $[\sin(nx)]/(nx)$ is a well-known mathematical function and is readily available in tabular form.

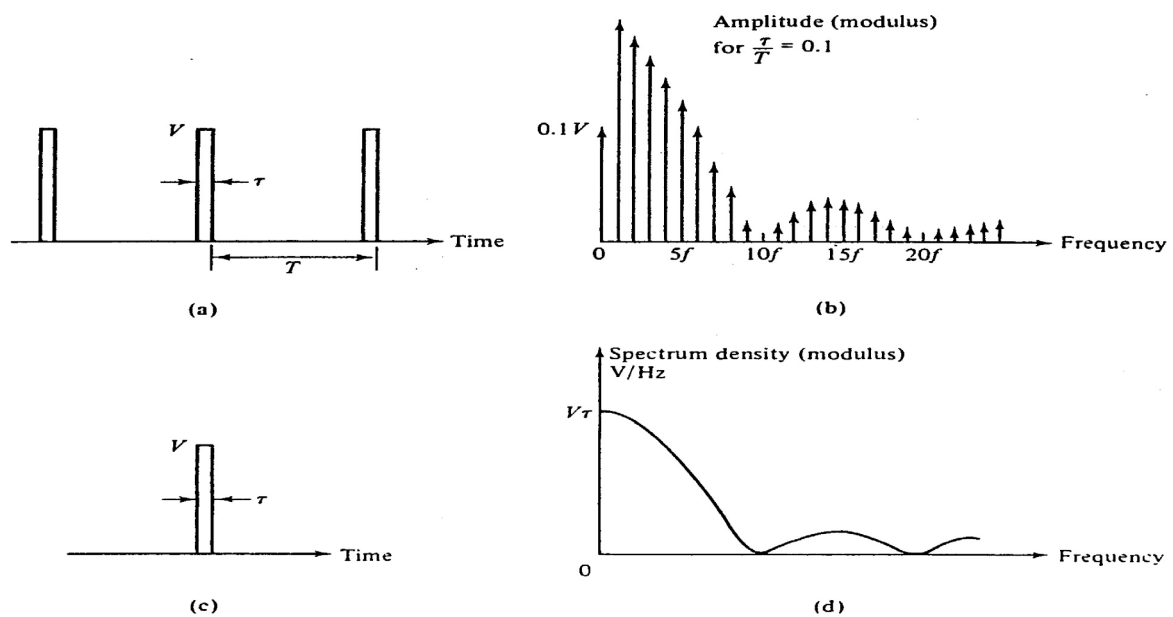


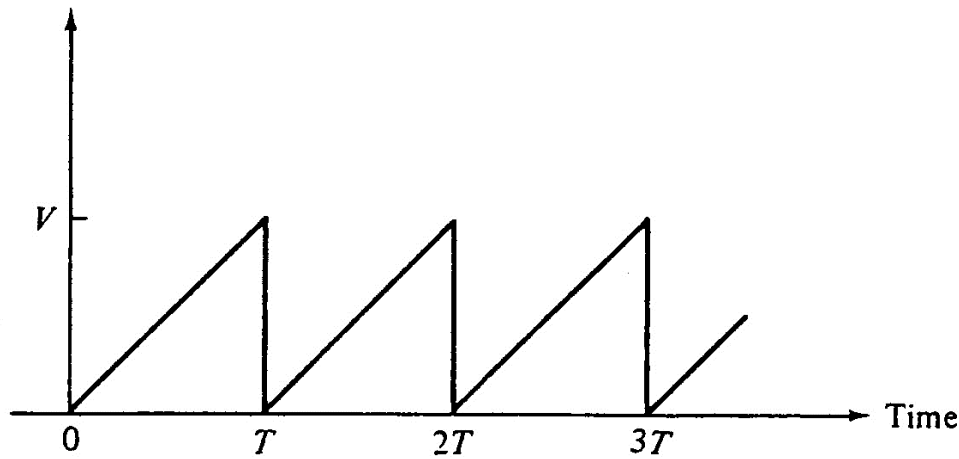
Figure 43: (a) A train of positive-going rectangular voltage pulses; (b) frequency spectrum; (c) a single positive-going rectangular voltage pulse; (d) spectrum density function of a single pulse.

The amplitude of the "nth" harmonic is seen to be

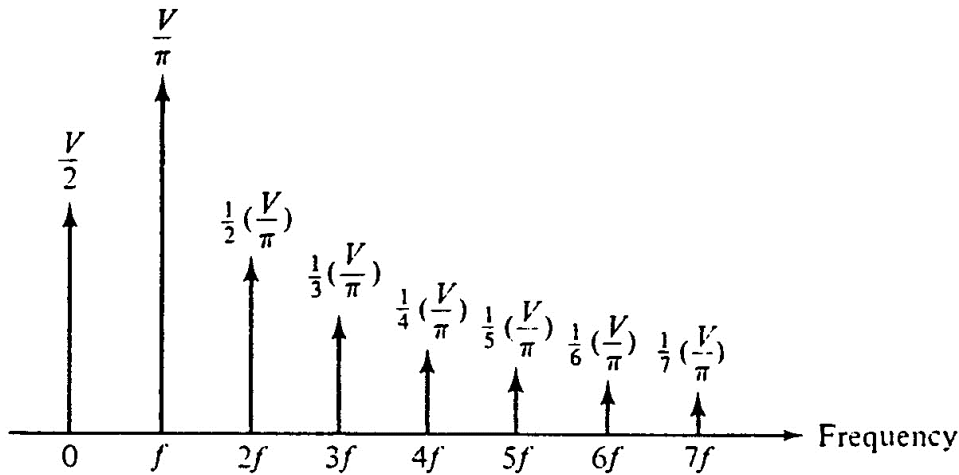
$$V_n = \frac{2V\tau}{T} \frac{\sin nx}{nx} \quad (13)$$

By letting T approach infinity, the train of rectangular pulses approaches a single pulse (Figure 28(c)). Also, as $T \rightarrow \infty$, $f \rightarrow 0$, and the spacing between the harmonic components shrinks to zero, so that the spectrum becomes *continuous* (Figure 28(d)). By continuous, it is meant that there are no discrete harmonic frequencies, and only the envelope of the spectrum can be determined. This is given by

$$V(f) = V\tau \frac{\sin \pi f \tau}{\pi f \tau} \quad (14)$$



(a)



(b)

Figure 44: (a) A rising-ramp triangular voltage waveform; (b) frequency spectrum.

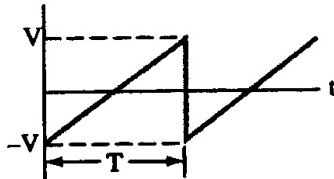
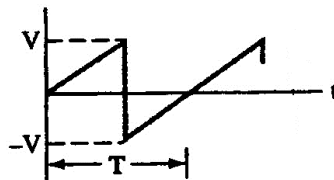
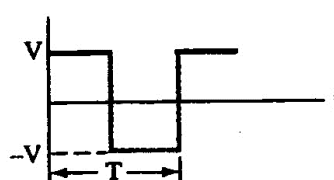
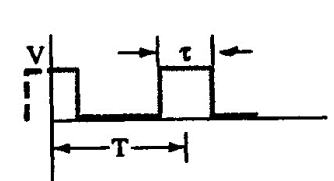
$V(f)$ in this case represents the *spectrum density* and has the units volts/frequency, i.e., V /Hz. The single pulse is an example of a non-repetitive wave and is included here because its properties follow easily from those of the repetitive pulse waveform.

Another function frequently encountered in practice is the sawtooth wave, shown in Figure 29(a). The Fourier series for this is

$$v = \frac{V}{2} - \frac{V}{\pi} \left(\sin \omega t + \frac{1}{2} \sin 2\omega t + \frac{1}{3} \sin 3\omega t + \dots \right) \quad (15)$$

The spectrum is shown in Figure 29(b).

Table II. Mathematical expressions of waveforms spectra

(a) 	$V = \frac{2V}{\pi} \left(\sin \omega t + \frac{1}{2} \sin 2\omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{4} \sin 4\omega t + \dots \right)$
(b) 	$V = \frac{2V}{\pi} \left(\sin \omega t - \frac{1}{2} \sin 2\omega t + \frac{1}{3} \sin 3\omega t - \frac{1}{4} \sin 4\omega t + \dots \right)$
(c) 	$V = \frac{4V}{\pi} \left(\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \dots \right)$
(d) 	$V = V \frac{\tau}{T} + 2V \frac{\tau}{T} \left[\frac{\sin \pi(\tau/T)}{\pi\tau T} \cos \omega t + \frac{\sin 2\pi(\tau/T)}{2\pi(\tau/T)} \cos 2\omega t + \frac{\sin 3\pi(\tau/T)}{3\pi(\tau/T)} \cos 3\omega t + \dots \right]$
(e) 	$V = \frac{8V}{\pi^2} \left[\cos \omega t + \frac{1}{(3)^2} \cos 3\omega t + \frac{1}{(5)^2} \cos 5\omega t + \dots \right]$
(f) 	$V = \frac{2V}{\pi} \left[1 + \frac{2 \cos 2\omega t}{3} - \frac{2 \cos 4\omega t}{15} + \dots (-1)^{n/2} \frac{2 \cos n\omega t}{n^2 - 1} \dots \right] (n \text{ even})$